

ACCENTURE INNOVATION CHALLENGE 2026 · PROBLEM TRACK 2 · ROUND 2

PatientTriage.ai

A decision-support layer over the emergency waiting room — sparse by design, continuously recomputed, and willing to say it does not know.

THE FROZEN CATEGORY

Acuity is assigned once at the door and never recomputed. Priority here is a trajectory, not a verdict.

ABSTENTION AS A FEATURE

The system refuses to guess between medium and high, and names the single question that would resolve the case.

ONE TABLET, OFFLINE

Stage 1 creates value from a complaint, five vitals and a visual check. No EHR, no labs, no identity, no internet.

TEAM

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DOCUMENT

Foundational project documentation

REVISION

1.0 · 23 August 2026

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A	AGENTS.md	Context file for AI coding agents
B	Decision Protocol	When a build agent decides, and when it escalates

These documents are written to be read in order and to be loaded, in full, as context for an AI coding agent before any code is generated. Each one closes a class of decision that would otherwise be improvised: the PRD fixes what is being built and why, the TRD fixes the stack and the scoring logic, the App Flow fixes every screen and trigger, the UI/UX Brief fixes every colour and size, the Backend Schema fixes the data contract, and the Implementation Plan fixes the order of work. Where a value is contestable – a clinical threshold, a cost ratio, a reassessment interval – it is named, sourced, and placed in a hospital-owned configuration file rather than buried in code.

00

Overview

Project documentation index

Team Mayhem · Ayush Kumar (lead), Md Waqar Moid, Harshit Agarwal · IIT Kanpur

Accenture Innovation Challenge 2026 · Problem Track 2 · Round 2

What this is

A decision-support layer over the emergency waiting room. It works from sparse arrival data — chief complaint, five vitals, age, sex and a nurse's visual check — recomputes risk as minutes pass, escalates on hard rules the hospital owns, refuses to guess when it cannot discriminate, and hands every final call back to the human.

It is not a diagnosis engine, an autonomous decision-maker, an EHR replacement, or a wait-time predictor.

The six documents

Read in order. Each one assumes the ones before it.

#	DOCUMENT	ANSWERS
1	Product Requirements	What we are building and why. Users, principles, functional requirements, metrics, rollout, risks, adoption, scalability.
2	Technical Requirements	The stack, the architecture, and the complete scoring-engine specification. Every prohibition an AI agent needs.
3	App Flow	Every screen, every navigation path, every interaction trigger. The five-minute demo path.
4	UI/UX Brief	The visual system. Colour tokens, type scale, spacing, components, motion, and the binding anti-generic contract.
5	Backend Schema	Data models, relations, constraints, authorisation, retention, and the privacy frame.
6	Implementation Plan	36 ordered build steps, each with an exit test. Definition of done.

Plus `AGENTS.md` — the short context file to load into Cursor, Claude Code, Windsurf or Lovable before generating any code — and `DECISION-PROTOCOL.md`, which governs what a build agent decides for itself and what it escalates.

The three claims the documents are built to support

- A frozen triage category is the hazard.** Priority here is a trajectory, not a verdict. `ENCOUNTER` has no `band` column; the current band is always the latest assessment.
- A system that cannot say "I don't know" is not safe.** Abstention is a first-class engine output with a schema constraint behind it, not a UI treatment on a low score.
- Stage 1 must work on one tablet, offline, in a district hospital at 2 a.m.** Every architectural decision — no build step, no framework, no server, no identity — follows from that.

Reading paths

IF YOU ARE	READ
A judge with ten minutes	PRD §1, §2, §8 · App Flow §13 (the demo path)
An AI coding agent	AGENTS.md first, then TRD and UI/UX Brief in full, then Implementation Plan step by step
A clinician	PRD §4 (principles) · TRD §4 (the scoring engine) · Backend Schema §6 (privacy and retention)
A designer	UI/UX Brief in full · App Flow §4, §5

Evidence base

- Sax DR, Warton EM, Mark DG, et al. *Evaluation of the Emergency Severity Index in US Emergency Departments for the Rate of Mistrriage*. JAMA Network Open, 2023. 5,315,176 encounters, 21 hospitals: 65.9% correct identification; 3.3% undertriage; 28.9% overtriage.
- *Emergency Severity Index Handbook*, 5th edition, 2023. Decision points A–D; high-risk vital thresholds; paediatric fever criteria.
- Royal College of Physicians. *National Early Warning Score 2 (NEWS2)*. Adult physiologic scoring tables.
- PALS / Maine EMS paediatric vital-sign reference ranges by age band.
- US FDA. *Clinical Decision Support Software: Guidance for Industry and FDA Staff*, revised January 2026. Basis for the recommend-not-decide boundary and the independent-review requirement.
- Academic College of Emergency Experts in India & WHO Collaborating Centre for Emergency and Trauma, South-East Asia. *Strategies to Combat Overcrowding at Emergency Departments across India*.

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01

Product Requirements

What we build, for whom, and how we know it worked

Team	Mayhem — Ayush Kumar (lead), Md Waqar Moid, Harshit Agarwal · IIT Kanpur
Programme	Accenture Innovation Challenge 2026 · Problem Track 2 · Round 2
Document	PRD — authoritative source for <i>what</i> is built and <i>why</i> . Supersedes any conflicting statement elsewhere.
Status	Baselined for the Round 2 prototype
Revision	1.0

1. The problem, stated precisely

In an overwhelmed emergency department one nurse sequences every arriving patient in under two minutes each, with no labs, no imaging, and often no history. **That judgement is then frozen.** A patient's position in the queue reflects how sick they looked on arrival, not how unsafe it has become for them to keep waiting.

The queue itself becomes the hazard.

Five mechanisms produce this failure, and the product must answer each one:

#	MECHANISM	CONSEQUENCE
M1	Snapshot assessment — acuity assigned once at the door, never recomputed	The patient who decompensates at minute 40 holds a minute-0 position
M2	Information poverty — the first five minutes contain vitals and a sentence, nothing that confirms or excludes	Decisions are made at the point of least evidence
M3	Presentations that lie — silent MI, afebrile sepsis, compensated paediatric shock	Look stable until they are not
M4	Crowding degrades reassessment — repeat vitals stop happening exactly when they matter	The safety net fails under the load that creates the need for it
M5	Single point of cognitive load — one nurse, continuous interruption, no memory of who is quietly worsening	Deterioration is discovered at collapse

Evidence base. Across 5,315,176 US ED encounters, only 65.9% of patients who went on to need a life-stabilising intervention were correctly identified at triage; 3.3% were undertriaged and 28.9% overtriaged under ESI ([Sax et al., JAMA Netw Open 2023](#)). At ESI decision point B — the high-risk judgement — measured nurse accuracy is approximately 43% ([ESI Handbook v5, 2023](#)). A high-volume Indian emergency department reported an average consultation delay of 14 hours in 2023.

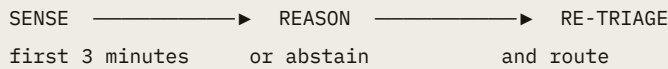
The gap is not nurse competence. It is that the task — continuous, multi-patient, multi-hour risk surveillance under interruption — is not a task a human can hold in working memory, and no conventional triage system attempts it.

2. What we are building

PatientTriage.ai is a decision-support layer over the waiting room, not a replacement for the triage nurse.

It works from sparse arrival data, recomputes risk as minutes pass, escalates on hard rules the hospital owns, refuses to guess when it cannot discriminate, and hands every final call back to the human.

Three verbs define the loop:



2.1 One-sentence product brief

A tablet-first, offline-capable board that keeps a live, continuously recomputed, explainable priority order for every person sitting in an emergency waiting room, using only chief complaint, five vitals, age, sex and a nurse's visual check — and that says "I don't know" out loud rather than guessing.

2.2 What it is not

- Not a diagnosis engine. It never names a disease.
- Not an autonomous decision-maker. It never assigns the final category, moves a patient, or orders treatment.
- Not an EHR replacement, and not dependent on one to deliver Stage-1 value.
- Not a wait-time predictor. Average wait is explicitly *not* an optimisation target.

3. Users

USER	CONTEXT	WHAT THEY NEED FROM US	FAILURE THEY FEAR
Triage nurse (primary)	Standing, interrupted every 90 seconds, 2 minutes per patient, no keyboard	Capture in under 90 seconds; a reason they can check in 3 seconds; zero-friction override	Being blamed for a machine's call
Charge nurse / ED physician	Running the floor, reallocating staff	Who is quietly worsening; who to see next; disaster-mode top five	Discovering a collapse in the chair
Hospital administrator / clinical governance	Weekly and quarterly review	Undertriage by subgroup; override rate; drift; a defensible audit trail	An inequitable system nobody noticed
The patient (indirect)	Sitting in a chair, deteriorating, invisible	To not be forgotten	Being sent to the back of the line

Design constraint that follows: the nurse's interaction budget is roughly **90 seconds of capture and 3 seconds of reading** per patient. Anything that costs more than that will not be used, and an unused safety system is worse than none because it produces a false record of vigilance.

4. Core principles (non-negotiable, and testable)

P	PRINCIPLE	TESTABLE FORM
P1	Hard rules outrank the model. On a visible emergency the model is locked out of ever advising "wait".	For every input where a red-flag rule fires, output priority = P1 and <code>model_locked_out = true</code> , regardless of model score.
P2	The queue is alive. A priority number is never frozen once set.	For every waiting patient, <code>last_recompute_age_seconds</code> $\leq$ 60.
P3	It says "I don't know." No score is returned without a confidence band; near a safety-relevant boundary it abstains and names the one resolving question.	No API response contains <code>band</code> without <code>confidence</code> and <code>interval</code> .
P4	The nurse decides. Override is one gesture, no dialog, no justification field. Every override is logged.	Override completes in $\leq$ 1 interaction and writes an immutable audit record.
P5	Undertriage is not symmetric with overtriage. Ties break upward.	Cost matrix is explicit and displayed; <code>undertriage_weight</code> / <code>overtriage_weight</code> = 8.
P6	Fairness is a running check, not a launch-day claim.	Bias monitor recomputes per shift and flags the <i>worst-served</i> subgroup, never the average.
P7	Front-door data only. Value is created from chief complaint, five vitals, age, sex and visual assessment alone.	Stage 1 runs with zero external integrations, offline, on one tablet.
P8	Local protocol supplies the hard rules. Red-flag thresholds are governed by the deploying hospital.	Rules live in an editable, versioned protocol file; changing it requires no code change.

P1 and P3 exist for a regulatory reason as well as a clinical one. The January 2026 FDA guidance on Clinical Decision Support Software keeps non-device status contingent on the clinician being able to **independently review the basis** for a recommendation, and treats time-critical decision support as failing that test ([Covington analysis, Jan 2026](#)). Our answer is to expose the full derivation of every score on the same screen, and to never present a single unqualified number.

5. Functional requirements

Priority key: **M** = must for Round 2 prototype · **S** = should · **C** = could (roadmap).

5.1 Sense — arrival capture

ID	REQUIREMENT	PRI
FR-1.1	Capture chief complaint as free text plus an optional complaint class	M
FR-1.2	Capture age and biological sex; accept "estimated" age for unidentified patients	M
FR-1.3	Capture HR, systolic BP, respiratory rate, SpO ₂ , temperature; each individually markable unobtainable	M
FR-1.4	Capture nurse visual checks as toggles: pale, diaphoretic, drowsy, distressed, cyanosed, active bleeding, unable to speak in full sentences	M
FR-1.5	Capture ACVPU level of consciousness	M
FR-1.6	Accept a patient with no identity, no history and no name — an encounter ID is generated and the record is complete without it	M
FR-1.7	Complete a full capture in ≤90 seconds on a tablet with no keyboard	M
FR-1.8	Record arrival mode (walk-in, ambulance, referred, police)	S
FR-1.9	Capture pain score 0–10 where the patient can give one	S

5.2 Reason — scoring and abstention

ID	REQUIREMENT	PRI
FR-2.1	Evaluate the hard-rule gate before any model computation; a fired rule pins the patient to P1 and locks the model out of any lower recommendation	M
FR-2.2	Compute an age-banded physiologic derangement score (adult NEWS2 tables; paediatric age-band tables; obstetric shifted thresholds)	M
FR-2.3	Compute a presentation-risk contribution from complaint class modified by age and sex — this is where atypical presentation is caught	M
FR-2.4	Emit a continuous Priority Index 0–100 and a band P1–P5 aligned to ESI levels 1–5	M
FR-2.5	Emit a confidence interval over bands , widened by every missing or unobtainable input	M
FR-2.6	Abstain when the interval spans a safety-relevant boundary: return UNRESOLVED with the two candidate bands	M
FR-2.7	On abstention, name the single most informative question for that complaint class	M
FR-2.8	Break every tie upward, applying an explicit asymmetric cost matrix, and state on screen that it did so	M
FR-2.9	Produce a complete, human-readable derivation for every score: every rule evaluated, every contribution, every point	M
FR-2.10	Never return a band without a confidence value — enforced at the engine boundary, not the UI	M
FR-2.11	Degrade gracefully to visual-assessment-only scoring when instruments are unavailable, widening the interval and naming the bands it can no longer discriminate	M

5.3 Re-triage — the living queue

ID	REQUIREMENT	PRI
FR-3.1	Recompute every waiting patient at least once per minute	M
FR-3.2	Apply a time-hazard contribution: minutes waited weighted by current band (a P2 waiting 30 minutes accrues risk far faster than a P4)	M
FR-3.3	Apply a drift contribution from the slope of repeated vitals over the observation window	M
FR-3.4	Raise a patient's position automatically when drift or hazard crosses threshold, and mark the row as movement with its cause	M
FR-3.5	Trigger mandatory re-assessment prompts when a band-specific reassessment interval elapses (P2 = 15 min, P3 = 30 min, P4 = 60 min, P5 = 120 min)	M
FR-3.6	Never allow a patient to be silently dropped from surveillance — an un-reassessed patient escalates in visibility, not out of it	M

5.4 Nurse authority and audit

ID	REQUIREMENT	PRI
FR-4.1	Any row can be moved to any position in one gesture. No dialog. No mandatory justification field.	M
FR-4.2	Every override writes an append-only audit record capturing the full system state at that instant: recommended band, confidence, all inputs, every rule fired, the nurse's resulting band, and timestamp	M
FR-4.3	An optional one-tap reason chip may be attached after the fact; it is never required before the move takes effect	M
FR-4.4	The audit log is exportable and readable by a clinician without the application	M
FR-4.5	Override rate and override direction are computed per shift and surfaced as a model-calibration signal — the model learns from the nurse, never the reverse	S

5.5 Safety modes

ID	REQUIREMENT	PRI
FR-5.1	Surge / disaster mode activates at $\geq 3\times$ baseline arrival rate: only the top five critical patients stay highlighted, all other rows collapse to a compact line, and the mode and its trigger are stated on screen	M
FR-5.2	Degraded-instrument mode when monitors are unavailable: fall back to visual checks, widen confidence, keep the queue moving	M
FR-5.3	Absolute emergency override fires a red alert to clinicians immediately and independently of queue position	M
FR-5.4	Every mode change is logged with its trigger	M

5.6 Fairness monitor

ID	REQUIREMENT	PRI
FR-6.1	Track assigned priority and subsequent acuity upgrades by sex, age band and language	M
FR-6.2	Report the worst-served subgroup , not the aggregate	M
FR-6.3	Flag drift when a subgroup's undertriage proxy diverges beyond a configured tolerance	M
FR-6.4	Present the monitor to administrators in a form they can act on, with the underlying encounters reachable	S

5.7 Overflow valve

ID	REQUIREMENT	PRI
FR-7.1	Where the department is at capacity, offer P4/P5 patients a guaranteed slot at a partner clinic	S
FR-7.2	Read live capacity across a city hospital network and recommend ambulance redirection when adding one more patient is itself the hazard	C

6. Non-functional requirements

ID	REQUIREMENT	TARGET
NFR-1	Scoring latency, single patient	< 10 ms
NFR-2	Full board recompute, 60 patients	< 100 ms
NFR-3	Cold load on a mid-range tablet	< 2 s
NFR-4	Offline operation — full function with no network for an entire shift, after a single initial load (service-worker cache) or from a packaged WebView	Mandatory (Stage 1)
NFR-5	Legibility at 1 m viewing distance, standing	Body ≥ 15 px, vitals ≥ 17 px, tabular figures
NFR-6	Colour is never the sole carrier of meaning	WCAG 1.4.1; every state also has a text token and a glyph
NFR-7	Contrast	≥ 4.5:1 body, ≥ 7:1 alarm states
NFR-8	Determinism — identical inputs produce an identical, reproducible score with a recorded engine version	Mandatory
NFR-9	No patient identifier is required, transmitted, or stored in Stage 1	Mandatory
NFR-10	Audit log is append-only and tamper-evident	Mandatory
NFR-11	Touch targets	≥ 44 × 44 px

7. Reference operating parameters

Stated as assumptions, per the Round 2 brief, and adjustable without code change.

PARAMETER	ASSUMED VALUE	WHERE IT IS USED
Department volume	100–500+ patient visits per day	Surge threshold calibration
Baseline arrival rate	6 patients/hour (mid-size ED)	Surge trigger = $\geq 3 \times = 18/\text{hour}$
Triage scale	5-level, ESI v5 aligned (P1–P5 ↔ ESI 1–5)	Band mapping
Prior-record availability	~50% of arrivals have some prior record; the system assumes none	Zero-history path is the default path, not the exception (§7.1)
Regulatory frame	DPDP Act 2023 (India) primary; designed to satisfy HIPAA §164 and GDPR Art. 9 by construction	Retention, consent, minimisation
Data retention	Encounter record 24 h on device; audit log 7 years, de-identified	Data policy
Undertriage : overtriage cost	8 : 1	Tie-breaking, threshold placement
Reassessment intervals	P2 15 min · P3 30 min · P4 60 min · P5 120 min	Re-triage prompts

These are directional, not fixed. Every one is a named constant in a single configuration file so that a deploying hospital changes its own policy without touching the engine.

7.1 Designing for the half with no record

The brief assumes roughly half of arrivals have a prior record. We design as though none of them do, and that is a deliberate choice rather than a simplification.

A system that performs well on the half with history and degrades on the half without has optimised for the easier population. The patient with no record is disproportionately the one who is unaccompanied, unable to speak, new to the area, or brought in by a stranger — and disproportionately the one who is undertriaged. Building the zero-history path as the *normal* path means the hard case is the one that is well tested, and the system cannot quietly become worse for the people it is already worst for.

It also makes the deployment claim honest. If value depended on record availability, Stage 1 could not be a single tablet in a department with no integration, and the pilot could not start until the integration did.

What the other half gets. History does not go unused; it arrives at Stage 2 as a *source of observations*, not a separate scoring path. An HL7/FHIR feed writes into `OBSERVATION` with `source = monitor_feed`, and prior conditions populate `preexisting_flags`, which existing presentation modifiers already read — anticoagulated, diabetic, cardiac history. The engine does not branch on whether a record exists. A patient with history simply has higher evidence completeness, and therefore a narrower interval, which is exactly the effect it should have.

8. Success metrics

We deliberately measure three things, and deliberately refuse to measure a fourth.

8.1 Catching the quiet crashes (*primary*)

What we count: patients sent to the back of the line who then deteriorated while sitting unmonitored. **Why:** this is the exact failure mode that kills people in a crowded department, and no conventional triage system measures it at all. **Pilot target:** a measurable fall in unplanned acuity upgrades and waiting-room collapses.

8.2 Speed for the sickest

What we count: P95 time-to-clinician for the highest-risk patients only. **Why:** averages hide the danger. A department can post a healthy mean wait while its sickest patients wait longest. **Pilot target:** faster first contact for the top-risk group **with the overtriage budget held fixed**, so the gain is real and not bought by upgrading everybody.

8.3 Equality of care

What we count: undertriage rate broken out by sex, age group and language. **Why:** pain is described differently across cultures and genders, and triage systems have been shown to absorb that difference as bias. **Pilot target:** no group with the same symptoms waits longer than another; the **worst-served** group is the optimisation target.

8.4 What we refuse to measure

Average waiting time. It is the metric that makes a department look good while its sickest patients wait longest. Optimising it directly conflicts with 8.2.

9. Round 2 prototype scope

The prototype is a **simulation harness around the real engine**. The scoring engine, rule gate, abstention logic, audit log and fairness monitor are production logic. The patients, the clock and the arrivals are synthetic.

In scope — and demonstrable:

#	ROUND 2 EXPECTATION	HOW THE PROTOTYPE SATISFIES IT
1	Triage scoring on 15–20 simulated records	20 synthetic encounters with scripted physiological trajectories
2	At least one ambiguous presentation	PT-0007 — 61 M, "stomach ache", diaphoretic, atypical ACS pattern
3	At least one paediatric case	PT-0011 — 3 y, T 38.5 °C, compensated shock physiology
4	At least one geriatric case	PT-0004 — 82 F, afebrile sepsis, normal-looking vitals
5	At least one zero-history / first-time patient	PT-0013 — unidentified, unresponsive, brought by passer-by
6	Behaviour under 3× surge	Surge control injects 18 arrivals/hour; board switches to disaster mode
7	Uncertainty surfaced on every score	No band renders without a confidence band; abstention is a first-class state
8	A clinician override, and the log it produces	Drag-to-top on any row; audit drawer shows the full captured state

Out of scope for Round 2 (explicitly, and stated in the pitch): EHR/HL7 integration · real patient data · authentication and role management · trained ML weights (the model layer is a calibrated rule-and-weight system, not a fitted classifier) · regulatory submission artefacts.

The Overflow Valve (FR-7.1, FR-7.2) is deliberately not prototyped, and this is worth stating rather than leaving as an absence. It needs a live capacity feed from partner clinics and a city hospital network — infrastructure that does not exist to integrate against, and that cannot be honestly simulated without inventing the one thing the feature depends on. Prototyping it would have produced a convincing screen backed by nothing.

That is also the argument the staged rollout makes: a hospital gets the whole of Stage 1 — a live, re-triaging, abstaining, auditable queue — before any of that infrastructure exists. A feature that requires the city to cooperate is the right thing to build third, and the wrong thing to demo first.

10. Rollout

Each stage delivers safety value on its own. A hospital does not need to reach stage three, or own a modern electronic record system, to get the benefit of stage one. Nothing here requires a large infrastructure upgrade to begin.

STAGE	NAME	WHAT CHANGES	WHAT IT NEEDS
Day 1	The Offline Tablet	Runs standalone on one tablet. The nurse types the basic numbers; hidden risk is flagged immediately. Keeps working when the hospital's internet goes down.	One tablet. Nothing else.
Month 3	The Background Listener	Connects securely to hospital records. The waiting-room list updates itself — the second a new blood pressure or lab result is saved, that patient's place in line adjusts.	Read-only HL7/FHIR feed
Future	The Traffic Director	Knows which beds and theatres are free. "This person is sick, Room 402 is empty." Or: "we are full, reroute this ambulance."	Bed-management and city-network integration

11. Risks and how the design answers them

RISK	ANSWER BUILT INTO THE PRODUCT
Automation bias — the nurse defers to the machine	The system abstains visibly and often; no unqualified number is ever shown; override is frictionless while agreement is not rewarded
Alert fatigue	Hard-rule alerts are rare by construction (rules are absolute, not probabilistic) and are the only interruptive element in the interface
Liability	Recommend-not-decide boundary; complete audit trail; clinician can independently review the basis of every recommendation on the same screen
Model drift	Fairness monitor plus override-rate calibration report; the engine version is recorded on every score
Workflow rejection	90-second capture budget, no mandatory fields at override, works offline, no new hardware
A rule set copied blindly between hospitals	Rules are a hospital-owned versioned artefact; the protocol version is displayed in the board header at all times
Under-resourced sites cannot deploy	Stage 1 is one tablet, offline, no integration, no identity requirement

12. Adoption — getting a fatigued staff to use it rather than work around it

A triage tool that is not used is worse than no tool, because it produces a record of vigilance that did not happen. The people who will decide whether this survives are at hour nine of a twelve-hour shift, interrupted every ninety seconds, and have been handed software before that made their night worse.

Five decisions in this product exist for adoption rather than for accuracy:

DECISION	WHAT IT BUYS
The override has no dialog and no justification field	The moment a nurse decides the machine is wrong is the moment a confirmation box becomes an insult. The gesture is one drag. The audit log, not the dialog, is what makes it safe.
The system abstains out loud	A tool that is confidently wrong once is never trusted again. A tool that says "I cannot separate P2 from P3, ask this" is a colleague. Abstention is the single largest trust investment in the design, and it costs accuracy on paper to buy credibility in use.
Alerts are rare by construction	Hard rules are absolute physiological criteria, not probabilistic thresholds, so they fire seldom and mean something when they do. The interruptive alert bar is the only interruptive element in the entire interface.
90-second capture, no new hardware, works offline	The tool asks for no workflow change that the department has to fund, schedule, or train around. It runs on a tablet they already have, in a corridor with no signal.
The nurse is never scored	Override rate is reported per shift and per cohort as a <i>model</i> calibration signal. It is never surfaced as an individual performance metric, and STAFF holds a pseudonymous ID with no name. A system that grades its users gets gamed, then abandoned.

Rollout as change management. Stage 1 changes nothing about how the department works: the nurse triages exactly as before, and the board runs alongside as a second opinion that cannot move anyone. The only thing it adds is that patients who are quietly worsening become visible. That is a small enough ask to say yes to, and it is the stage at which trust is either earned or not.

What we would measure in a pilot. Override rate and direction in week one versus week six; the proportion of abstentions that get an answer rather than being ignored; and the number of shifts where the board is left closed. The last one is the real adoption metric, and it is the one most pilots do not collect.

13. Scalability — one assistant across very different hospitals

A workflow built for a large urban trauma centre does not transfer to a small rural department, and a scoring model tuned on one case mix is unsafe on another. The product handles this by making almost nothing about a hospital a property of the software.

Clinical variation lives in a file, not in code. `protocol.v1.json` holds every contestable value: red-flag rules and their thresholds, age bands, physiologic tables, presentation classes and modifiers, band thresholds, hazard rates, reassessment intervals, the cost ratio, and the surge trigger. A paediatric hospital ships a protocol whose population paths are weighted differently. An obstetric unit extends the obstetric rules. A rural department with a two-bed resuscitation area raises its own surge threshold, because for them three times baseline is a different number. None of that is a software release, and the protocol version is stamped on every assessment so a decision stays interpretable under the rules that produced it.

Scale variation is a parameter, not an architecture. The surge trigger is a multiple of *that department's* measured baseline arrival rate, not an absolute count — so a department seeing 100 visits a day and one seeing 500 both enter surge when they are three times their own normal, which is the only definition of "overwhelmed" that means the same thing in both places.

Technical-maturity variation is the rollout. A hospital with no electronic record, no network in the triage corridor, and one spare tablet gets the whole of Stage 1. A hospital with a modern EHR adds the Stage 2 listener and the queue starts updating itself. A hospital with bed management and a regional network reaches Stage 3. Each stage delivers safety value alone, and — the point that matters for adoption — **a hospital never has to reach the next stage to keep the value of the last one.**

What does not flex, and should not. The engine, the invariants, the audit contract and the recommend-not-decide boundary are the same everywhere. A hospital may change what counts as a red flag; it may not change whether a fired red flag can be overruled by the model, whether a score can be returned without a confidence value, or whether an override is logged. Those are the properties that make the thing safe, and a deployment that could switch them off would not be the same product.

14. Open questions

1. Who owns the protocol file clinically — ED clinical lead, or hospital quality committee? Affects the change-control workflow.
2. Should override reason chips be configurable per department, or standardised for cross-site comparability?
3. What is the minimum pilot duration for the fairness monitor to produce a statistically defensible subgroup signal at 200 visits/day?
4. In degraded-instrument mode, is there a floor below which the system should refuse to score at all rather than score wide?

Sources: [Sax DR et al., JAMA Netw Open 2023](#) · [ESI Handbook 5th Edition, 2023](#) · [Royal College of Physicians NEWS2](#) · [FDA CDS Software guidance, revised Jan 2026](#)

02

Technical Requirements

Stack, architecture, scoring-engine specification

Governs	Every technical decision. If an AI coding agent is about to guess a library, framework, file location or algorithm, the answer is here.
Depends on	01-PRD.md
Revision	1.0

1. The one hard constraint

No build step. No bundler. No framework. No node_modules in the deploy path.

The files that live in the repository are the files that are served. Nothing is compiled, transpiled, bundled or generated. The identical tree deploys to Vercel by pointing Vercel at the GitHub repository with zero configuration, and runs on a tablet with no internet.

What "runs offline" means precisely. Browsers block ES modules and service workers on the `file://` scheme, so a bare double-click of `index.html` is *not* the supported path. Offline operation is delivered two ways, and both are first-class:

1. **Service-worker cache.** The application is loaded once over HTTPS — from Vercel, or from any local server on the hospital network — after which `sw.js` serves every asset from cache. The device can then be offline indefinitely, across reloads and reboots. This is the Stage-1 deployment.

2. **Packaged WebView.** The same unmodified files wrapped in an Android WebView, where the `file://` restriction does not apply. This is the route for a tablet that will never see a network at all.

The distinction that matters for the product claim is **no build step and no server-side runtime**, not the `file://` scheme specifically. A developer inspecting the tree locally uses `python3 -m http.server 8000` (§10).

This is not a shortcut. It is the Stage-1 product requirement (PRD NFR-4, FR-7 assumption "front-door data only, one tablet") expressed as an architecture. A build step is a dependency on a laptop, a Node version and a network — three things a district hospital at 2 a.m. does not have.

Consequences, accepted deliberately:

- Native ES modules (`<script type="module">`), served over HTTP(S) or `file://` with a documented local-server fallback.
- No JSX, no TypeScript compilation, no PostCSS, no Tailwind.
- No npm runtime dependencies. Dev dependencies exist only for testing and are never shipped.

2. Stack — decided, not to be re-litigated

LAYER	DECISION	WHY THIS AND NOT THE OBVIOUS ALTERNATIVE
Markup	Hand-written semantic HTML5, one page	A framework would add 40 kB and a build step to render a table
Styling	One hand-written CSS file. Custom properties for the token layer. CSS Grid + Flexbox.	Explicitly not Tailwind. Tailwind's defaults are the single largest source of the generic AI-generated look (see 04-UIUX-BRIEF.md §2). A hand-written sheet forces every value to be chosen.
Behaviour	Vanilla ES2022 modules. No React, Vue, Svelte, Alpine, htmx, jQuery.	The whole application is one list, one detail panel and a timer. A framework here is cost without benefit, and it would put a build step between us and the tablet.
State	A single observable store module (<code>state.js</code>) with an explicit reducer-style <code>dispatch</code> . No proxies, no magic.	Auditable state transitions matter for a clinical audit log. Magic reactivity does not.
Rendering	Targeted DOM patching in <code>render.js</code> . Full re-render of the board is permitted (60 rows in <16 ms); rows are keyed by encounter ID and reused.	Virtual DOM is unnecessary at n=60. Measure before optimising.
Persistence	<code>localStorage</code> for session state, <code>IndexedDB</code> for the append-only audit log. Both wrapped in <code>storage.js</code> , both fully optional — the app runs with storage unavailable.	Stage 1 is device-local by design; nothing needs a server.
Offline	A service worker (<code>sw.js</code>) with a cache-first strategy over a precache manifest.	This is the "keeps working when the internet goes down" claim, made real.
Charts	Hand-authored inline SVG generated in JS. No Chart.js, D3, Plotly, Recharts.	Two charts (bias bars, trend sparklines). A charting library would be 60 kB to draw thirty rectangles, and would import its own visual defaults.
Fonts	IBM Plex Sans, IBM Plex Mono, IBM Plex Serif via Google Fonts, self-cached by the service worker, with a full system fallback stack.	Deliberate non-default type. Never Inter, Roboto, Poppins, Montserrat or Space Grotesk.
Icons	Inline SVG, 1 px stroke, hand-drawn, five glyphs total. No Lucide, Feather, Heroicons, Font Awesome, or emoji.	Icon libraries carry the visual fingerprint we are avoiding.
Testing	Node's built-in <code>node:test</code> runner for the engine. Playwright for the board smoke test. Both dev-only.	Zero runtime dependency; the engine is pure functions and trivially testable.
Hosting	Vercel, static, from GitHub main. <code>vercel.json</code> sets headers only.	Zero-config static deploy is the point.
Runtime deps	None.	

2.1 Explicit prohibitions for any coding agent working on this repo

Do not introduce, suggest, or scaffold: React · Next.js · Vite · Tailwind CSS · shadcn/ui · Bootstrap · Chart.js · D3 · Lucide/Feather/Heroicons · Google Material · any CSS-in-JS · any state library · any date library (use `Intl` and `Date`) · any HTTP client (use `fetch`) · any polyfill for a feature supported in Chrome/Safari/Firefox since 2022.

If a task appears to require one of these, the task has been misread. Re-read this document.

3. Repository layout

This is the required application tree, not an exclusion list. Files that support the project without being served — documentation, the decision log, tests, `.gitignore` — belong in the repository and are shown below. Nothing here may be deleted to "match the tree"; anything not listed and not obviously supporting needs a decision-log entry before it is added.

The **deployed surface** is exactly: `index.html`, `sw.js`, `manifest.webmanifest`, `vercel.json`, and `assets/`. `docs/` and `tests/` ship in the repository but are never fetched by the application.

```

patienttriage-ai/
├─ index.html          Board. The only HTML entry point.
├─ sw.js               Service worker. Precache + cache-first.
├─ vercel.json         Headers + clean URLs. No build config.
├─ manifest.webmanifest Installable on a tablet home screen.
├─ .gitignore          Test and tooling artefacts only (§3.1).
├─ README.md
├─ PROMPT-FOR-CODEX.md Build brief. Not served.
├─ docs/
│   ├─ 00-README.md    Index and reading paths
│   ├─ 01-PRD.md
│   ├─ 02-TRD.md        (this file)
│   ├─ 03-APP-FLOW.md
│   ├─ 04-UIUX-BRIEF.md
│   ├─ 05-BACKEND-SCHEMA.md
│   ├─ 06-IMPLEMENTATION-PLAN.md
│   ├─ 07-DECISION-LOG.md Appended to as the build proceeds
│   ├─ AGENTS.md        Context file for coding agents
│   └─ PatientTriage-ai_Project-Documentation.pdf
├─ tests/
│   ├─ unit/           node:test – one file per engine module
│   │   ├─ physiology.test.js NEWS2 tables, paediatric bands
│   │   ├─ rules.test.js   Every hard rule, positive and negative
│   │   ├─ presentation.test.js
│   │   ├─ hazard.test.js
│   │   ├─ uncertainty.test.js
│   │   └─ bands.test.js
│   ├─ property/
│   │   └─ invariants.test.js 10,000 random encounters (§9)
│   ├─ golden/
│   │   ├─ cohort.test.js    Snapshot runner
│   │   └─ snapshots/        Committed golden files, t=0/30/60 min
│   ├─ e2e/
│   │   └─ board.spec.js     Playwright
│   ├─ fixtures/
│   │   └─ news2-chart.json  The published chart, transcribed
│   └─ package.json         devDependencies ONLY. Never deployed.
└─ assets/
    ├─ css/
    │   └─ board.css         Single stylesheet. Tokens at the top.
    ├─ data/
    │   ├─ protocol.v1.json  HOSPITAL-OWNED. Hard rules + thresholds.
    │   └─ cohort.json       20 synthetic encounters + trajectories.
    └─ js/
        ├─ main.js           Entry. Wires everything. No logic.
        ├─ state.js          Store, dispatch, subscribe.
        ├─ clock.js          Simulation clock, tick loop, speed control.
        └─ engine/
            ├─ index.js       score(encounter, protocol, now) → Assessment
            ├─ rules.js        Layer 0 – hard-rule gate
            ├─ physiology.js   Layer 1 – age-banded derangement score
            └─ presentation.js Layer 2 – complaint × age × sex risk

```

	hazard.js	Layer 3 – waiting-time + vital drift
	uncertainty.js	Confidence interval, abstention, resolving question
	bands.js	Index → band mapping, asymmetric tie-break
	sim/	
	cohort.js	Loads cohort, advances trajectories
	surge.js	3× arrival injection, disaster mode
	audit.js	Append-only log, IndexedDB, export
	fairness.js	Subgroup undertriage proxy + worst-served
	render/	
	board.js	Queue rail
	inspector.js	Right-hand derivation panel
	modes.js	Surge / degraded banners
	charts.js	Inline SVG: sparkline, bias bars, confidence band
	util/	
	dom.js	el(), on(), text() – 30 lines, no library
	fmt.js	Time, deltas, tabular numbers
	storage.js	localStorage + IndexedDB wrappers

3.1 .gitignore

Test and tooling artefacts only. The application has no build output, so there is nothing else to ignore — an inherited Node `.gitignore` template here would be noise.

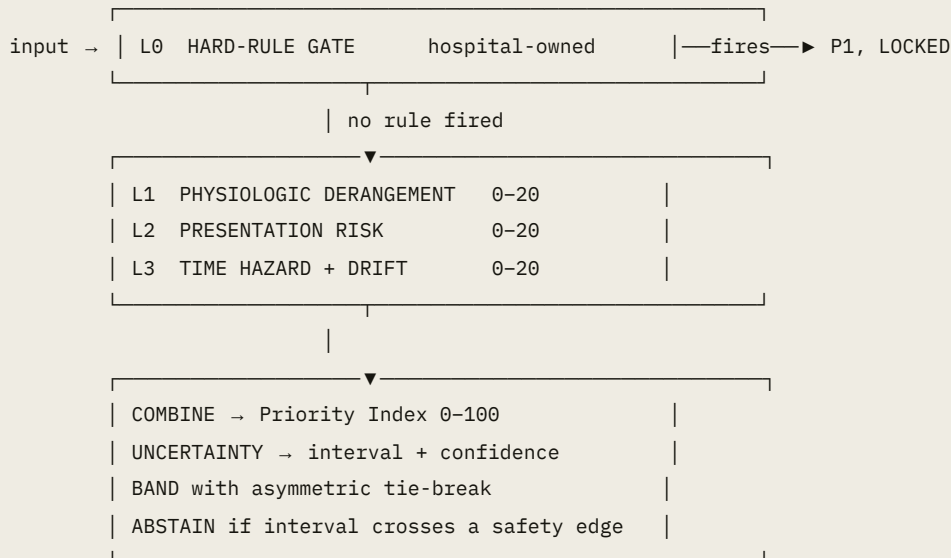
```
node_modules/
tests/e2e/.playwright/
tests/e2e/test-results/
tests/e2e/playwright-report/
coverage/
.DS_Store
Thumbs.db
.vercel/
```

`tests/package.json` and its lockfile **are** committed — the test toolchain must be reproducible. Only `node_modules/` is ignored, and it lives under `tests/`, never at the repository root, so that no tooling mistakes this for a Node application.

Rule for agents: one module, one responsibility. `engine/` is pure — no DOM access, no `Date.now()`, no storage. `now` is always passed in. This is what makes the engine deterministic (PRD NFR-8) and testable.

4. Scoring engine specification

The engine is **four layers evaluated in a fixed order**. It is deliberately not one opaque model. A nurse must be able to read the derivation in three seconds, and a regulator must be able to read it at all.



4.1 Layer 0 — Hard-rule gate (engine/rules.js)

Deterministic, hospital-owned, evaluated first, **cannot be outvoted by any model output**.

Each rule is a data object in `protocol.v1.json` :

```

{
  "id": "RULE-RESP-01",
  "label": "Critical hypoxaemia",
  "population": "adult",          // adult | paediatric | obstetric | all
  "condition": { "field": "spo2", "op": "<", "value": 85 },
  "action": "PIN_P1",            // PIN_P1 | FLOOR_P2 | FLAG
  "alert": "immediate",
  "rationale": "NEWS2 SpO2 score 3 with additional margin; airway/oxygenation emergency.",
  "source": "RCP NEWS2 2017; local protocol v1"
}
  
```

Baseline rule set shipped in the prototype (a deploying hospital replaces it):

ID	POPULATION	CONDITION	ACTION
RULE-AIRWAY-01	all	visual.airway_compromise = true	PIN_P1
RULE-CONSC-01	all	acvpv ∈ {P, U}	PIN_P1
RULE-CONSC-02	all	acvpv ∈ {C, V}	FLOOR_P2
RULE-BLEED-01	all	visual.active_major_bleeding = true	PIN_P1
RULE-RESP-01	adult	spo2 < 85	PIN_P1
RULE-RESP-02	adult	rr ≥ 30 or rr ≤ 8	PIN_P1
RULE-CIRC-01	adult	sbp < 90 and (hr > 120 or visual.pale)	PIN_P1
RULE-CIRC-02	adult	hr ≥ 131 or hr ≤ 30	FLOOR_P2
RULE-PAED-01	paediatric	age_days < 90 and (temp ≥ 38.0 or temp < 36.0)	FLOOR_P2
RULE-PAED-02	paediatric	sbp < 70 + 2 × age_years (age 1–10)	PIN_P1
RULE-PAED-03	paediatric	hr or rr above the age-band ceiling in §4.2.2	FLOOR_P2
RULE-PAED-04	paediatric	spo2 < 90	PIN_P1
RULE-PAED-05	paediatric	rr at or below the age-band bradypnoea floor	PIN_P1
RULE-OBS-01	obstetric	pregnant and (sbp < 90 or sbp > 150)	FLOOR_P2
RULE-OBS-02	obstetric	pregnant and heavy_vaginal_bleeding	PIN_P1
RULE-SEIZ-01	all	visual.seizure_active = true	PIN_P1

PIN_P1 sets band = P1, model_locked_out = true, alert = immediate. When model_locked_out is true the engine may not emit any recommendation containing a wait. This is asserted in the engine, not the UI (PRD P1).

Rule sources are ESI v5 decision points A and B, and the NEWS2 single-parameter-3 trigger.

4.2 Layer 1 – Physiologic derangement (engine/physiology.js)

Age-banded. Returns { score, perParameter[], missing[] }.

4.2.1 ADULT (≥ 16 Y) – NEWS2, UNMODIFIED

PARAMETER	3	2	1	0	1	2	3
Respiration /min	≤8		9–11	12–20		21–24	≥25
SpO ₂ % (scale 1)	≤91	92–93	94–95	≥96			
Air or oxygen		O ₂		Air			
Systolic BP mmHg	≤90	91–100	101–110	111–219			≥220
Pulse /min	≤40		41–50	51–90	91–110	111–130	≥131
Consciousness				Alert			CVPU
Temperature °C	≤35.0		35.1–36.0	36.1–38.0	38.1–39.0	≥39.1	

Aggregate bands: 0 routine · 1–4 low · 5–6 medium (urgent) · ≥7 high (emergency). Any single parameter scoring 3 is itself a trigger, independent of aggregate — implement this as a separate returned flag, not as an aggregate comparison.

Source: [Royal College of Physicians NEWS2 chart](#).

4.2.2 PAEDIATRIC (< 16 Y) – AGE-BAND TABLE

NEWS2 is adult-calibrated and dangerous in children. Use age-band normals; score the *deviation from band*, not from an adult constant.

AGE BAND	HR NORMAL	RR NORMAL	SBP NORMAL	HYPOTENSION FLOOR
Neonate < 28 d	100–205	30–60	67–84	< 60
Infant 1 mo–1 y	100–190	30–53	72–104	< 70
Toddler 1–2 y	98–140	22–37	86–106	< 70 + 2×age
Preschool 3–5 y	80–120	20–28	89–112	< 70 + 2×age
School-age 6–11 y	75–118	18–25	97–115	< 70 + 2×age
Adolescent 12–15 y	60–100	12–20	110–131	< 90

ESI v5 paediatric high-risk ceilings, used by RULE-PAED-03: HR >190 (<1 mo), >180 (1–12 mo), >140 (1–3 y), >120 (3–12 y), >100 (12–18 y); RR >60, >55, >40, >35, >30, >20 respectively.

Scoring for these three: 0 within band; 1 if within 15% outside; 2 if 15–30% outside; 3 if >30% outside or beyond an ESI ceiling.

The other four parameters. Layer 1 scores seven parameters in every population, and all three paths sum to exactly 20 — that is where `layerMax: 20` comes from. HR, RR and SBP use the deviation table above. The remaining four:

PARAMETER	PAEDIATRIC HANDLING	MAX	WHY
SpO₂	Own table, shifted one notch more sensitive than adult: $\leq 91 \rightarrow 3$ · $92-94 \rightarrow 2$ · $95-96 \rightarrow 1$ · $\geq 97 \rightarrow 0$	3	Paediatric room-air normal is ≥ 97 , and a child holds saturation by respiratory effort until that effort fails. The same reading carries more risk than in an adult.
Supplemental oxygen	Identical to adult: air 0, oxygen 2	2	The oxygen score exists because supplemental O ₂ masks the SpO ₂ reading. That masking is age-independent, so a separate table would be unjustified precision.
ACVPU	Numerically identical to adult: A 0, C/V/P/U 3	3	What differs is the <i>assessment</i> , not the score. Under 1 year, Alert is judged behaviourally — recognises and fixes on the carer, consolable, normal tone. A quiet, floppy or inconsolable infant is not Alert. RULE-CONSC-01/02 are already <code>population: all</code> .
Temperature	<code>adultNEWS2.temp_c</code> unchanged	3	The thermoregulatory set point does not vary meaningfully by age. The <i>urgency</i> does, and that is carried by RULE-PAED-01 and the fever presentation modifiers, not by the score.

Two paediatric hard rules exist because the adult ones are unsafe to reuse. RULE-RESP-01 (`spo2 < 85`) and RULE-RESP-02 (`rr ≤ 8`) are `population: adult` and must never be applied to a child.

- **RULE-PAED-04** pins at `spo2 < 90` — *above* the adult threshold, not below it, because a desaturating child is closer to arrest than an adult at the same reading.
- **RULE-PAED-05** pins on bradypnoea below an age-band floor (neonate 21 · infant 21 · toddler 15 · preschool 14 · school 13 · adolescent 8, each ~30% below the lower normal bound). A falling respiratory rate in a child is exhaustion, not improvement. Without this rule a 3-year-old at RR 10 — normal range 20–28 — would score as merely deranged rather than peri-arrest, because the adult constant of 8 is meaningless at that age.

Stated limitation. A child with known congenital cyanotic heart disease has a documented baseline saturation that the SpO₂ table will overscore. Stage 1 holds no prior record and cannot detect that case. The nurse override is the mitigation, and the limitation is written into the protocol file rather than left implicit.

Critical implementation note. A fever of 38.5 °C carries entirely different urgency in a 3-year-old than in a 75-year-old, and a solution applying one adult-calibrated table across all ages introduces a silent safety risk. The age band is selected *before* any table lookup, never after.

Sources: [PALS/Maine EMS paediatric vital signs reference](#), [ESI Handbook v5](#).

4.2.3 OBSTETRIC

Adult table with shifted circulatory thresholds. Systolic follows ACOG rather than the adult table: 141–150 scores 1 (gestational hypertension begins at 141, which the adult table scores 0), 151–160 scores 2, ≥161 scores 3, 90–99 scores 2, ≤89 scores 3. HR > 110 scores 2 (physiological pregnancy tachycardia and lower baseline BP make the adult table both over- and under-sensitive in the wrong directions).

4.3 Layer 2 — Presentation risk (`engine/presentation.js`)

This layer exists to catch **presentations that lie**. It maps the chief complaint to a class, then applies age and sex modifiers.

```
presentationRisk = base[complaintClass]
                  + Σ modifiers(age, sex, pregnancy, visual, arrivalMode)
```

COMPLAINT CLASS	BASE	NOTABLE MODIFIERS
chest_pain	12	+4 if diabetic or age ≥ 65; +3 if female (atypical ACS under-recognition); +4 if diaphoretic; +3 if exertional onset
abdominal_pain	6	+7 if age ≥ 55 and (diaphoretic or radiating) — the "stomach ache" that is an MI; +5 if pregnant
breathlessness	11	+3 if unable to speak full sentences; +3 if age ≥ 70
altered_behaviour	10	+4 if age ≥ 70 (delirium as sepsis presentation); +4 if age < 5
headache	5	+10 if sudden/worst-ever (thunderclap) — reaches the P2 floor alone, per ESI decision point B; +5 if with fever and neck stiffness; +8 if pregnant or postpartum (pre-eclampsia until excluded)
fever	4	+9 if age < 90 days; +5 if age ≥ 65 (afebrile/atypical sepsis — ESI uses 65 as the geriatric cutoff)
weakness_nonspecific	5	+6 if age ≥ 75 — the most under-weighted complaint in geriatric ED presentation
trauma_minor	3	+7 if anticoagulated — reaches the P3 floor alone; +5 if age ≥ 75 (falls, occult head injury)
limb_injury	2	+4 if neurovascular compromise reported
minor_illness	1	—
unknown	8	Used for unidentified/non-communicative patients. Absence of a complaint is not absence of risk.

Every modifier that fires is returned in the derivation with its label and points. Nothing contributes invisibly.

Presentation band floors — and why they exist.

PRESENTATION SCORE	FLOOR
≥ 15	P2
≥ 10	P3

This is **ESI v5 decision point B expressed as data**: a high-risk presentation is level 2 *regardless of vital signs*. It is applied exactly like a `FLOOR_P2` rule (§12.4 step 5) — it raises the final band and never lowers it.

It is not a tuning convenience. Without it the continuous weighting makes the product's central claim impossible to deliver, and the arithmetic is worth stating because it is not obvious:


```
weighted sum = 2.2·L1 + 1.6·L2 + 1.0·L3          max 96
P2 requires  PI ≥ 62  →  weighted sum ≥ 59.5
```

A patient with normal physiology (L1 = 1), maximum presentation risk (L2 = 20)
and maximum accrued hazard (L3 = 20):
2.2 + 32 + 20 = 54.2 → PI 56.5 → P3 at best.

A patient with normal vital signs could never exceed P3, however dangerous the presentation. That makes silent MI, afebrile sepsis and compensated paediatric shock — the three failure modes named in the PRD as the reason this product exists — architecturally uncatchable. The continuous score is the right instrument for *how deranged is this physiology*; it is the wrong instrument for *is this presentation dangerous*, which is a threshold judgement, and ESI already models it as one.

bandSetBy . Every assessment records which mechanism decided the band: `model` · `hard_rule` · `presentation_floor` · `nurse_override` . The inspector shows it, because "P2 because a rule fired" and "P2 because the index landed there" are different claims and a clinician reviewing the decision needs to know which one they are reading.

Clamping. `score = min(base + Σ firing modifiers, 20)` . Only `chest_pain` can exceed the ceiling (max 26). When the sum is clamped, **the derivation must still list every modifier that fired and mark the result as clamped**. A nurse reading the reasoning must see everything the system saw, not only what fitted inside the ceiling — a hidden contribution is indistinguishable from a contribution that was never made.

On PM-AB-01. The atypical-ACS modifier fires on `age ≥ 55 AND (diaphoretic OR radiating)` . Both limbs matter and neither is redundant: radiating pain in an over-55 with an abdominal complaint is the textbook silent-MI presentation, and diaphoresis is the one the patient will not report. Requiring both together would mean a 61-year-old describing pain spreading to the left arm scored no higher than one with simple gastritis, which is exactly the failure this layer exists to prevent.

4.4 Layer 3 — Time hazard and drift (`engine/hazard.js`)

This is what makes the queue alive rather than a sorted snapshot.

```
timeHazard = waitedMinutes × hazardRate[currentBand]
drift       = Σ over parameters of slope(parameter, window) × sensitivity[parameter]
layer3      = clamp(timeHazard + drift, 0, 20)
```

BAND	HAZARD RATE (INDEX POINTS / MIN)	REASSESSMENT INTERVAL
P1	— (already highest)	continuous
P2	0.22	15 min
P3	0.10	30 min
P4	0.035	60 min
P5	0.012	120 min

Drift sensitivities (per unit change over a 30-minute window): RR +1.4 · SpO₂ −1.8 · HR +0.5 · SBP −0.6 · temperature +0.4. Drift is computed over the observed series only; a single reading yields drift = 0 and *widens the confidence interval* instead of contributing zero silently.

Worked example (the demo case). PT-0002, 34 F, breathlessness. Arrival RR 22 → 45 minutes later RR 27, SpO₂ 96 → 93. Drift = (5/30 × 1.4 × 30) + (3/30 × 1.8 × 30) = 7.0 + 5.4 = 12.4 → clamped contribution moves the index across the P3/P2 boundary. The row rises, and the movement cause reads `RR ↑5 · SpO2 ↓3 over 45 min` .

4.5 Combination and banding (engine/bands.js)

$$\text{PriorityIndex} = 100 \times (2.2 \cdot L1 + 1.6 \cdot L2 + 1.0 \cdot L3) / (2.2 \cdot 20 + 1.6 \cdot 20 + 1.0 \cdot 20)$$

Weights encode that measured physiology outranks reported complaint, which outranks elapsed time.

BAND	INDEX RANGE	ESI EQUIVALENT
P1	≥ 82 , or any PIN_P1 rule	ESI 1
P2	62 – 81	ESI 2
P3	38 – 61	ESI 3
P4	18 – 37	ESI 4
P5	< 18	ESI 5

Asymmetric tie-break. With the cost matrix below, the expected-cost-minimising decision at an ambiguous boundary is the *higher* acuity. When the confidence interval straddles a boundary and the mass split is closer than 65:35, the engine takes the higher band and sets `tie_broken_upward = true`, which the interface states in words.

	TRUE HIGHER ACUITY	TRUE LOWER ACUITY
Assigned higher	0	1 (overtriage)
Assigned lower	8 (undertriage)	0

4.6 Uncertainty and abstention (engine/uncertainty.js)

Evidence completeness $E \in [0,1]$:

$$E = \Sigma \text{weight}(\text{available inputs}) / \Sigma \text{weight}(\text{all inputs})$$

Weights: HR 0.14 · SBP 0.14 · RR 0.16 · SpO₂ 0.16 · temp 0.08 · ACVPU 0.12 · complaint 0.12 · visual checks 0.08.

Interval half-width on the Priority Index:

$$\begin{aligned} \text{halfWidth} &= \text{baseWidth} \times (1 + 2.2 \times (1 - E)) + \text{driftUncertainty} + \text{singleReadingPenalty} \\ \text{baseWidth} &= 5.0 \\ \text{singleReadingPenalty} &= 6.0 \text{ when only one observation set exists} \end{aligned}$$

driftUncertainty . A slope is only as trustworthy as the number of readings behind it and the span they cover. A large drift measured from two readings six minutes apart, then extrapolated across a thirty-minute window, is a guess; the same drift from five readings spanning the full window is a measurement. The formula says so:

```

driftUncertainty = 0.35 × |layer3DriftContribution|
                  × sparsityFactor × coverageFactor × consistencyFactor

sparsityFactor    = 2 / (nObservations - 1)          n=2 → 2.0 · n=3 → 1.0 · n=5 → 0.5
coverageFactor    = clamp(driftWindowMinutes / observedSpanMinutes, 1.0, 2.0)
consistencyFactor = 1 - 0.65 × monotonicFraction      range [0.35, 1.0]

driftUncertainty = 0 when nObservations < 2          (singleReadingPenalty applies instead)
capped at 8.0

```

monotonicFraction is, for each parameter contributing to drift, the fraction of consecutive steps whose direction matches that parameter's overall slope — averaged across contributing parameters, weighted by the size of each contribution. It is **0** when **nObservations < 3**: two points cannot agree or disagree, so they carry no consistency evidence.

Why the consistency term exists. An earlier revision scaled uncertainty with drift magnitude alone, on the reasoning that rapid change makes a point estimate unreliable. That reasoning is half right, and the missing half is damaging: it made the engine **least certain about the patients deteriorating fastest** — the exact cases the product exists to catch. In the reference cohort it pushed PT-0002 and PT-0010, the two most clearly worsening patients, into abstention at the moment they crossed into P2.

Magnitude is not the only signal. Four readings moving the same way across three parameters is an established trajectory, and the system should say so. The same magnitude inferred from two noisy readings is a guess. Consistency is what separates them, and without it the formula punishes exactly the evidence that should increase confidence.

The original intuition survives in the two terms that earned it: **sparsity** and **coverage**. A large drift from few readings over a short span is still wide.

Confidence is reported as a label, never as a false-precision percentage. Five states:

CONFIDENCE	CONDITION	ABSTAINS	INSTRUCTION TO THE NURSE
ESTABLISHED	Interval lies wholly inside one band	no	—
PROBABLE	Interval crosses one boundary, split ≥ 65:35	no	—
UNRESOLVED	Interval crosses a boundary, split < 65:35, and a question exists that would resolve it	yes	Ask the named question
UNRESOLVABLE	Same ambiguity, but no question can resolve it	yes	Questioning will not help. Escalate for clinician review.
INSUFFICIENT	E < 0.45	yes	Obtain the missing observations

Abstention is reserved for safety-relevant boundaries: P1/P2 and P2/P3.

Below those, an interval that straddles a boundary tie-breaks upward and reports **PROBABLE**, with no resolving question. Uncertainty about whether a patient needs one resource or none is real, but it changes nothing about their safety, and spending the nurse's single question on it is worse than useless: **abstaining everywhere would make abstention noise, and noise is how a safety signal stops being read.** A board where a third of the rows say "I don't know" has not communicated honesty; it has communicated nothing.

This also explains why almost every patient reads **PROBABLE** at the door. One observation set carries a 6.0 width penalty, so arrival intervals are wide by construction. That is correct — the system genuinely knows less at minute zero — and it is why the queue exists.

UNRESOLVED and **UNRESOLVABLE** carry identical uncertainty. They are separate states because they give the human different instructions, and the instruction is the point — a nurse told "ask this" and a nurse told "asking won't help, get a

clinician" are being asked to do different things. Collapsing them into one state with a nullable question would hide that difference behind a null check.

`UNRESOLVABLE` is reached when, and only when, one of the three `noQuestionReasons` holds: every question for the class was suppressed by `alreadyAnsweredWhen`, no question clears `minimumInformationGain`, or the class defines no questions. The reason is recorded, not inferred.

Abstention output. When `UNRESOLVED` or `INSUFFICIENT`, the engine returns:

```
{
  "band": null,
  "candidateBands": ["P2", "P3"],
  "confidence": "UNRESOLVED",
  "provisionalBand": "P2",          // the safe upper choice, used for queue position
  "resolvingQuestionId": "RQ-ABDO-02",
  "resolvingQuestion": "Is the abdomen rigid, or is there rebound tenderness?",
  "expectedInformationGain": 0.77,
  "noQuestionReason": null
}
```

The patient is **queued at the provisional (higher) band** while unresolved. Abstention is never a reason to defer care.

Resolving-question selection. Per complaint class, `protocol.v1.json` holds a ranked list of discriminating questions with the band-shift each would produce. The engine picks the one with the largest expected shift across the current candidate bands. Exactly one question is shown. Asking a nurse three questions in a crowded department is the same as asking none.

A question is never asked about something already recorded. Each question may carry an `alreadyAnsweredWhen` condition, using the same grammar as the rules (§12.2). Where it evaluates true, the finding was captured at arrival and has *already* contributed through a presentation modifier — offering the question would double-count it and would spend the single question the nurse gets on information the system already holds. The engine skips it and offers the next-best question for the class.

Worked case: PT-0007 arrives with the `radiating` qualifier ticked. PM-AB-01 has already fired on it, so RQ-ABDO-01 ("does the pain radiate...") is suppressed and RQ-ABDO-02 ("is the abdomen rigid, or is there rebound tenderness?") is offered instead — a question that actually adds information.

expectedInformationGain. A question earns its place by landing its two outcomes on opposite sides of the boundary that separates the candidate bands — decisively, and far apart. Scored on 0–1:

```
boundary = bandThresholds[higher candidate band]
yesPI    = priorityIndex + expectedShiftIfYes
noPI     = priorityIndex + expectedShiftIfNo

separates = (yesPI ≥ boundary) XOR (noPI ≥ boundary)
margin    = min(|yesPI - boundary|, |noPI - boundary|)

EIG = 0.5 × (separates ? 1 : 0)
      + 0.3 × min(margin / halfWidth, 1)
      + 0.2 × min(|yesPI - noPI| / (2 × halfWidth), 1)
```

The three terms are, in order: *does the answer actually decide it* · *does it decide it cleanly rather than landing on the line* · *does it move the estimate enough to matter against the current uncertainty*. Separation dominates, because a question whose answers both fall on the same side of the boundary tells the nurse nothing regardless of how large the shift is.

Worked, on PT-0007. `priorityIndex` 63.4, interval [55.1, 71.7] so `halfWidth` 8.3, candidates P2/P3 so `boundary` 62.0. RQ-ABDO-01 is suppressed (`radiating` already recorded), leaving RQ-ABDO-02 with shifts +11 / -4:

```
yesPI = 74.4    noPI = 59.4
separates = (74.4 ≥ 62) XOR (59.4 ≥ 62) = true
margin    = min(12.4, 2.6) = 2.6

EIG = 0.5 × 1 + 0.3 × min(2.6 / 8.3, 1) + 0.2 × min(15.0 / 16.6, 1)
     = 0.5      + 0.3 × 0.3133              + 0.2 × 0.9036
     = 0.7747 → 0.77
```

Note that any separating question scores **at least 0.50** by construction. A value below 0.5 therefore means the question does not decide the case, and such a question should rarely survive ranking — if one does, the candidate set is wrong.

These numbers are executable. `protocol.v1.json` → `workedExamples` carries this case, the unsuppressed RQ-ABDO-01 variant (0.8663), and the `driftUncertainty` table as fixtures. The Step 10 and Step 11 tests assert against them, so a constant that drifts out of step with a formula fails a test rather than waiting to be spotted by eye. Any change to a formula or a weight must update the fixtures and keep them green.

Ranking is deterministic: highest EIG, then the question whose `discriminatesBetween` exactly equals `candidateBands`, then order of appearance in the protocol file. Never random, never dependent on object key order.

The floor. If the best available question scores below `minimumInformationGain` (0.25), it is not offered. Asking a question that will not change the answer costs the nurse the same twenty seconds as asking a useful one, and spends the credibility that makes them answer the next one. The encounter becomes `UNRESOLVABLE` with reason `no_question_above_information_threshold`.

If every question for the class was suppressed by `alreadyAnsweredWhen`, the reason is `all_questions_already_answered`. The engine never falls back to a question it has already answered, and never manufactures confidence it does not have.

4.7 Assessment object (engine output contract)

```
{
  "encounterId": "PT-0007",
  "engineVersion": "1.0.0",
  "protocolVersion": "v1",
  "computedAt": 1756000000000,
  "priorityIndex": 63.4,
  "interval": [55.1, 71.7],
  "band": null,
  "provisionalBand": "P2",
  "bandSetBy": "model",
  "candidateBands": ["P2", "P3"],
  "confidence": "UNRESOLVED",
  "evidenceCompleteness": 0.78,
  "modelLockedOut": false,
  "tieBrokenUpward": true,
  "rulesFired": [],
  "derivation": {
    "physiology": { "score": 6, "perParameter": [ ... ], "missing": ["temp"] },
    "presentation": { "class": "abdominal_pain", "base": 6,
      "modifiers": [{ "label": "Age ≥55 with diaphoresis", "points": 7 } ] },
    "hazard": { "waitedMinutes": 38, "timeHazard": 3.8,
      "drift": 2.1, "driftDetail": "HR ↑12 over 30 min" }
  },
  "resolvingQuestion": "Is the abdomen rigid, or is there rebound tenderness?",
  "resolvingQuestionId": "RQ-ABDO-02",
  "expectedInformationGain": 0.77,
  "noQuestionReason": null,
  "reassessDueAt": 1756000900000
}
```

Contract assertions, enforced in `engine/index.js`. The function throws if it is about to return an object where:

- `confidence` is absent
- `band` and `provisionalBand` are both null
- `priorityIndex` falls outside `interval`
- `confidence === 'UNRESOLVED'` and `resolvingQuestionId` is null
- `confidence === 'UNRESOLVABLE'` and `noQuestionReason` is null, or `resolvingQuestionId` is non-null
- `noQuestionReason` is set on any state other than `UNRESOLVABLE`
- `modelLockedOut` is true and `provisionalBand !== 'P1'`

These mirror the database constraints in Backend Schema §3.3 exactly, so the same violation is impossible at both ends. PRD FR-2.10 is a code-level invariant, not a UI convention.

5. Simulation harness

The prototype wraps the real engine in a synthetic environment. **No simulation code may live inside `engine/`.**

COMPONENT	BEHAVIOUR
sim/cohort.js	Loads 20 encounters from cohort.json, each with a scripted trajectory: an array of { atMinute, vitals, visual } keyframes, linearly interpolated.
clock.js	A simulation clock at 1× / 10× / 60× speed. Emits a tick; every tick advances trajectories and recomputes the whole board. Pausable. now is <i>always</i> taken from this clock, never Date.now().
sim/surge.js	Injects arrivals at 3× the 6/hour baseline (18/hour). Sets mode = SURGE when the trailing-15-minute arrival rate ≥ 3× baseline.
Degraded mode	A control that marks all instrument-derived vitals unobtainable, forcing the visual-only path.

Cohort composition (20 encounters), each mapped to a requirement it proves:

ID	PATIENT	PROVES
PT-0001	47 M, chest pain, diaphoretic	Baseline P2, established confidence
PT-0002	34 F, breathlessness, worsening RR	Live drift re-triage (§4.4 worked example)
PT-0003	29 M, ankle injury	P5 floor, fast-track
PT-0004	82 F, "just not right", afebrile sepsis	Geriatric atypical presentation; normal-looking vitals, high presentation risk
PT-0005	55 M, headache, sudden onset	Thunderclap modifier
PT-0006	23 F, 32 weeks pregnant, SBP 148	Obstetric threshold shift
PT-0007	61 M, "stomach ache", diaphoretic	Ambiguous case → abstention + resolving question
PT-0008	71 M, fall, on warfarin	Anticoagulation modifier
PT-0009	40 F, migraine, known	P4 stable
PT-0010	66 M, cough + fever 38.2	P3 → drifts to P2
PT-0011	3 y, T 38.5, HR 168, cap refill 3 s	Paediatric age-band scoring; compensated shock
PT-0012	19 M, laceration	P4
PT-0013	Unidentified, unresponsive, no history	Zero-history path; PIN_P1 via RULE-CONSC-01
PT-0014	58 F, palpitations, HR 132	FLOOR_P2 rule
PT-0015	6 mo, poor feeding, T 37.8	Infant band; low absolute numbers, high relative deviation
PT-0016	45 M, back pain	P4, long waiter — time hazard accrual
PT-0017	77 F, confusion, new	ACVPU C → FLOOR_P2
PT-0018	31 M, hand burn	P4
PT-0019	88 M, weakness, no monitor available	Degraded-instrument mode; INSUFFICIENT confidence
PT-0020	52 F, chest pain, anxious, normal vitals	Sex modifier; the case the system must not dismiss

6. Data, privacy and security

CONCERN	STAGE 1 DECISION
Patient identity	Not required. Encounter IDs are generated locally, sequential per shift, meaningless outside the device.
PHI at rest	Vitals + complaint only. No name, no MRN, no address, no phone.
PHI in transit	None. Stage 1 makes zero network calls after the initial asset load.
Storage	<code>localStorage</code> (session) and IndexedDB (audit), both device-local. Cleared on shift close except the audit log.
Retention	Encounter records 24 h; audit records 7 years, de-identified at shift close (age band retained, exact age dropped).
Regulatory frame	DPDP Act 2023 (India) as primary; the minimisation design also satisfies HIPAA §164.312 and GDPR Art. 9(2)(h) without change.
Consent	Stage 1 processes no identified personal data, so it operates on the care-provision basis; the consent artefact becomes relevant at Stage 2 (EHR link) and is specified there.
Audit integrity	Append-only. Each record carries a SHA-256 hash chained to its predecessor (<code>prevHash</code>), making silent modification detectable. Implemented with <code>crypto.subtle</code> .
Transport	Vercel serves HTTPS only; <code>Strict-Transport-Security</code> , <code>X-Content-Type-Options: nosniff</code> , and a CSP with no <code>unsafe-inline</code> script are set in <code>vercel.json</code> .

7. Performance budget

METRIC	BUDGET
Total transferred (cold, gzipped)	≤ 120 kB including fonts — this is the governing budget

Why the per-asset budgets are gzipped. Earlier revisions capped uncompressed source bytes. That measures something no device ever downloads, and a ceiling fitted to the code as written blocks the next legitimate change by a few hundred bytes while the metric that matters still has kilobytes of headroom — which is exactly what happened when the resolving-question short labels were added. The budgets now measure transferred bytes, and the total transfer remains the governing constraint. | JS, uncompressed | ≤ 135 kB | | CSS, **gzipped** | ≤ 8 kB | | Cold load to interactive, mid-range tablet | ≤ 2 s | | Full board recompute, 60 encounters | ≤ 100 ms | | Single-patient score | ≤ 10 ms | | Tick interval | 1 s wall clock | | Long tasks during a tick | 0 over 50 ms |

8. Browser and device targets

Primary	Chrome/Chromium 110+ on a 10" Android tablet, landscape, 1280×800
Secondary	Safari 16.4+ (iPad), Firefox 115+, desktop Chrome/Edge
Minimum viewport	1024×768 (below this the inspector becomes a drawer)
Input	Touch first; full keyboard operation required for accessibility
Not supported	Internet Explorer; any browser without ES modules or <code>crypto.subtle</code>

9. Testing requirements

LEVEL	SCOPE	TOOL
Unit	Every engine function. NEWS2 table exactness against the published chart. Paediatric band boundaries. Hypotension formula at ages 1, 5, 10.	node:test
Property	For 10,000 random encounters: no output ever lacks confidence; PIN_P1 always yields P1; the interval always contains the point estimate.	node:test
Regression	Golden-file snapshot of all 20 cohort assessments at t=0, t=30, t=60 min.	node:test
Integration	Board renders 20 rows; drift moves PT-0002; surge switches mode; override writes an audit record and the hash chain verifies.	Playwright
Visual	Screenshots at 1280×800 and 1024×768, light and dark.	Playwright
Accessibility	Keyboard-only pass through capture → override → audit export. Contrast audit.	Manual + axe

Test files live under `tests/` per §3, are written as ES modules importing directly from `assets/js/engine/`, and are never referenced by `index.html`, `sw.js`, or the precache manifest.

```
cd tests && npm install          # devDependencies only: @playwright/test
node --test unit/ property/ golden/
npx playwright test e2e/
```

Deployment gate: no push to `main` without unit + property tests green.

10. Deployment

```
git init
git add -A
git commit -m "PatientTriage.ai – Round 2 prototype"
git remote add origin git@github.com:<org>/patienttriage-ai.git
git push -u origin main
```

On Vercel: **Add New** → **Project** → **import the repo** → **Framework Preset:** `other` → **Build Command:** `(empty)` → **Output Directory:** `.` → **Deploy**. There is no build step to configure. Every push to `main` redeploys; every pull request gets a preview URL.

`vercel.json`:

```
{
  "cleanUrls": true,
  "headers": [
    { "source": "/sw.js",
      "headers": [{ "key": "Cache-Control", "value": "no-cache" }] },
    { "source": "/assets/(.*)",
      "headers": [{ "key": "Cache-Control", "value": "public, max-age=31536000, immutable" }] },
    { "source": "/(.*)",
      "headers": [
        { "key": "X-Content-Type-Options", "value": "nosniff" },
        { "key": "Referrer-Policy", "value": "no-referrer" },
        { "key": "Strict-Transport-Security", "value": "max-age=63072000" }
      ] }
  ]
}
```

Local run. `python3 -m http.server 8000` or `npm serve .` — required only because ES modules and service workers are blocked on `file://`. A packaged tablet build wraps the same files in a WebView, where this restriction does not apply.

11. Third-party services

SERVICE	USED FOR	FALLBACK IF UNAVAILABLE
Google Fonts	IBM Plex Sans / Mono / Serif	Full system font stack; layout is unaffected because metrics are set on the fallback
Vercel	Static hosting, HTTPS, preview deploys	Any static host, or the raw files on a USB stick

That is the complete list. No analytics, no error reporting service, no CDN for scripts, no third-party API. A clinical tool that phones home is a clinical tool that leaks.

12. Protocol file schema

`assets/data/protocol.v1.json` is shipped complete in this repository. It is the **single source of every clinically contestable value** — thresholds, weights, rules, questions. No such value may appear as a literal in code.

This section documents its shape so the engine can be written against it without reading the whole file. Where §4 above and the JSON disagree, **the JSON wins** — it is the artefact the engine loads.

12.1 Top-level keys

KEY	CONSUMED BY	§4 REFERENCE
ageBands, ageBandFallback	engine/physiology.js → ageBand()	§4.2
rules	engine/rules.js	§4.1
physiology.adultNEWS2	engine/physiology.js	§4.2.1
physiology.paediatricNormals, .paediatricDeviationScoring, .paediatricHighRiskCeilings, .paediatricHypotensionFloor	engine/physiology.js	§4.2.2
physiology.obstetricOverrides	engine/physiology.js	§4.2.3
presentation.classes	engine/presentation.js	§4.3
hazard	engine/hazard.js	§4.4
uncertainty	engine/uncertainty.js	§4.6
combination, bandThresholds, costMatrix	engine/bands.js	§4.5
reassessMinutes, surge	clock.js, sim/surge.js	§4.4, App Flow §10
resolvingQuestions	engine/uncertainty.js	§4.6

12.2 Condition grammar

One recursive grammar, used by `rules[].condition`, `presentation.classes.*modifiers[].when`, and `resolvingQuestions[].alreadyAnsweredWhen`. Implement the evaluator **once**, in `engine/rules.js`, and import it into `engine/presentation.js` and `engine/uncertainty.js`.

```
condition := leaf | { "all": [condition, ...] } | { "any": [condition, ...] }
leaf      := { "field": <path>, "op": <operator>, "value": <literal>, "table": <path> }
```

`all` and `any` may nest to any depth. An empty `all` is true; an empty `any` is false.

OPERATOR	SEMANTICS
< <= > >= = !=	Numeric or string comparison against value
in	value is an array; true when the field is a member
contains	Field is an array; true when it contains value
isTrue	Boolean field is exactly true. No value key.
belowPaediatricHypotensionFloor	Resolves <code>physiology.paediatricHypotensionFloor</code> for the encounter's age band, evaluating expression where it is a formula. No value key.
belowPaediatricBradypnoeaFloor	Resolves <code>physiology.paediatricBradypnoeaFloor</code> for the encounter's age band and compares at-or-below. No value key.
aboveAgeBandCeiling	Looks up table (a dotted path under <code>physiology</code>) at the encounter's age band and compares strictly greater.

A leaf whose field is null, absent, or `unobtainable` evaluates to `false` — never to an error and never to `true`. A rule cannot fire on data that was not obtained. Missing data widens the interval instead (§4.6); it never manufactures a hard escalation.

12.3 Field paths

Resolved against a flat view of the encounter joined to its latest observation.

PATH	SOURCE
hr sbp dbp rr spo2 temp_c acvpu pain_score cap_refill_s	Latest OBSERVATION
visual.*	OBSERVATION.visual
age_days age_years age_band sex pregnancy_status gestation_weeks	ENCOUNTER, derived before evaluation
preexisting_flags complaint_class complaint_qualifiers	ENCOUNTER
population	Derived from age_band and pregnancy_status; obstetric outranks adult

`complaint_qualifiers` is a `TEXT[]` on `ENCOUNTER` set by the arrival-capture qualifier chips: `thunderclap` · `neck_stiffness` · `neurovascular_compromise` · `radiating` · `exertional` · `sudden_onset` · `reported_change_from_baseline`. It exists so that a presentation modifier can depend on something the nurse actually observed, rather than on free-text parsing — the engine never reads `complaint_text`.

12.4 Rule evaluation order

1. Filter `rules` to those whose `population` matches the encounter's population, or is `all`.
2. Evaluate every remaining rule. **Do not short-circuit** — the derivation must list every rule that fired, not the first.
3. Apply the strongest action: `PIN_P1` outranks `FL00R_P2` outranks `FLAG`.
4. `PIN_P1` sets `band = "P1"`, `modelLockedOut = true`, `alert = "immediate"`, and skips layers 1–3 for banding purposes. Layers are still computed and recorded for the derivation, because a reviewer must be able to see what the model would have said.
5. `FL00R_P2` computes the model band normally and then raises it to P2 if it came out lower. It never lowers a band.
6. The **presentation floor** (§4.3) is applied at the same stage and by the same logic. Where both a hard rule and a presentation floor apply, the higher band wins and `bandSetBy` records `hard_rule` — a fired rule is the more specific finding and the one a reviewer will want named.
7. A floor that raises the band above every candidate band **resolves an abstention**: the ambiguity was between bands the floor has now overruled. Confidence becomes `ESTABLISHED` with `bandSetBy` naming the floor. The engine does not report uncertainty about a decision it did not make on the model's evidence.

12.5 Scoring tables

Band arrays (`adultNEWS2.*`, `obstetricOverrides.*`) are lists of `{ min?, max?, score }` with **inclusive** bounds. A missing `min` means unbounded below, a missing `max` unbounded above. Bands are contiguous and non-overlapping; the evaluator returns the first match and must throw if no band matches, since that means the table has a gap.

`paediatricNormals` gives `[low, high]` per parameter per age band, covering **HR, RR and SBP only**. `paediatricDeviationScoring` maps percentage distance outside that range to a score; `maxPercentOutsideBand: null` is the catch-all. Exceeding a `paediatricHighRiskCeilings` value scores 3 regardless of the percentage.

The other four Layer-1 parameters take their paediatric values from `paediatricSpo2.bands`, `paediatricOnOxygen`, `paediatricAcvpu` and `adultNEWS2.temp_c` (§4.2.2).

`physiology.parameterMaxima` is the contract that makes this checkable: it lists the per-parameter maximum, the total of 20, and — under `populationPaths` — the exact table each population reads for each of the seven parameters. **Assert against it in the Step 7 unit tests**. If a population path cannot reach 20, a parameter is missing from that path, which is precisely the class of gap that leaves a child unscored on something an adult is scored on.

12.6 Changing the protocol

A hospital edits this file through its own clinical governance. Increment `protocolVersion`, set `supersedes` to the previous version, and keep the superseded file — an assessment must remain interpretable under the rules in force when it was made (Backend Schema §6). The version renders in the board header at all times and is stamped on every assessment.

Sources: [RCP NEWS2 chart](#) · [ESI Handbook v5, 2023](#) · [Maine EMS paediatric vital signs reference](#) · [FDA CDS guidance analysis, Jan 2026](#) · [Vercel builds documentation](#)

03

App Flow

Screens, paths, interaction triggers

Purpose	Removes all guesswork about what exists on screen and what happens when it is touched. If a behaviour is not in this document, it does not exist.
Depends on	01-PRD.md, 02-TRD.md
Revision	1.0

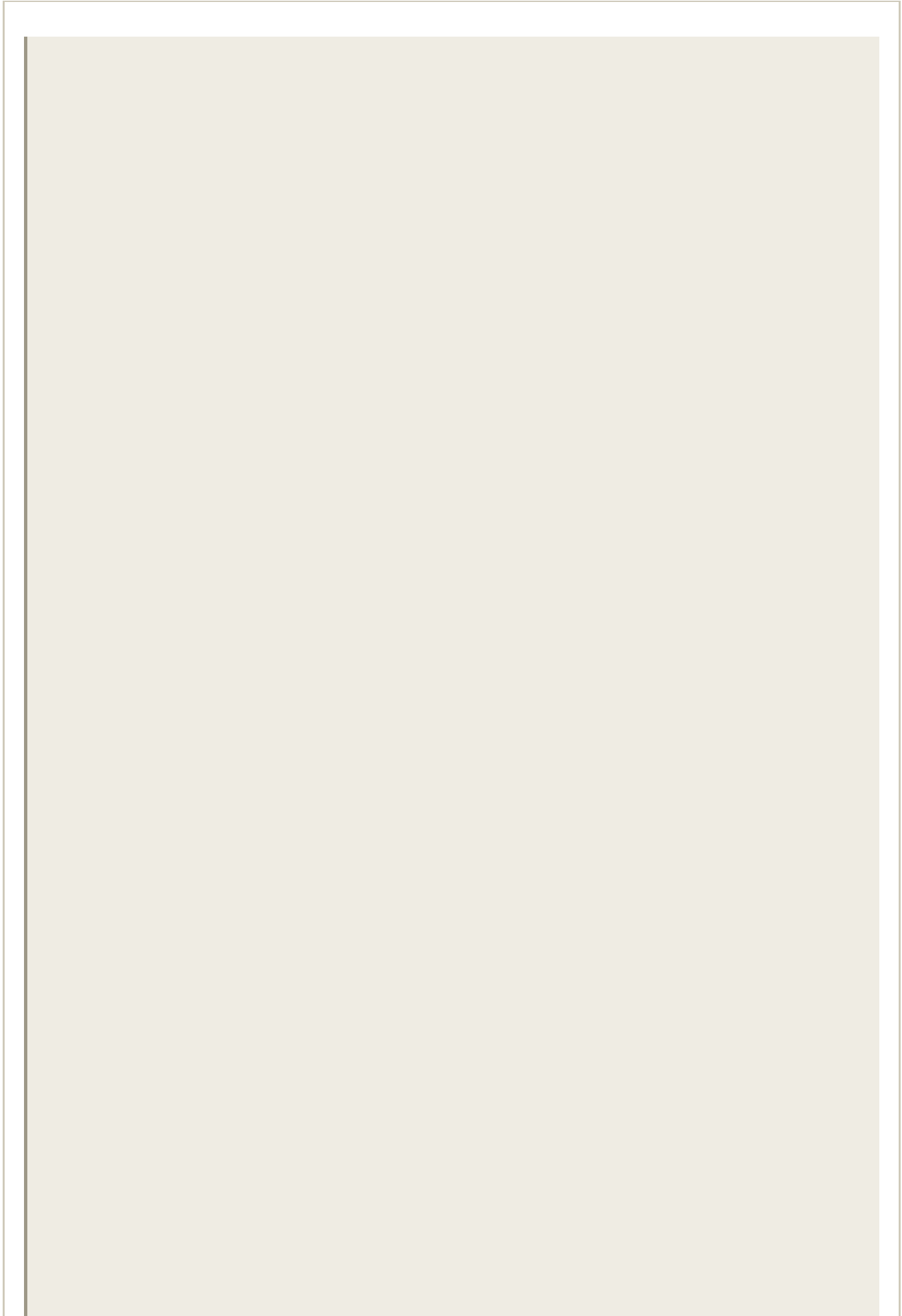
1. Screen inventory

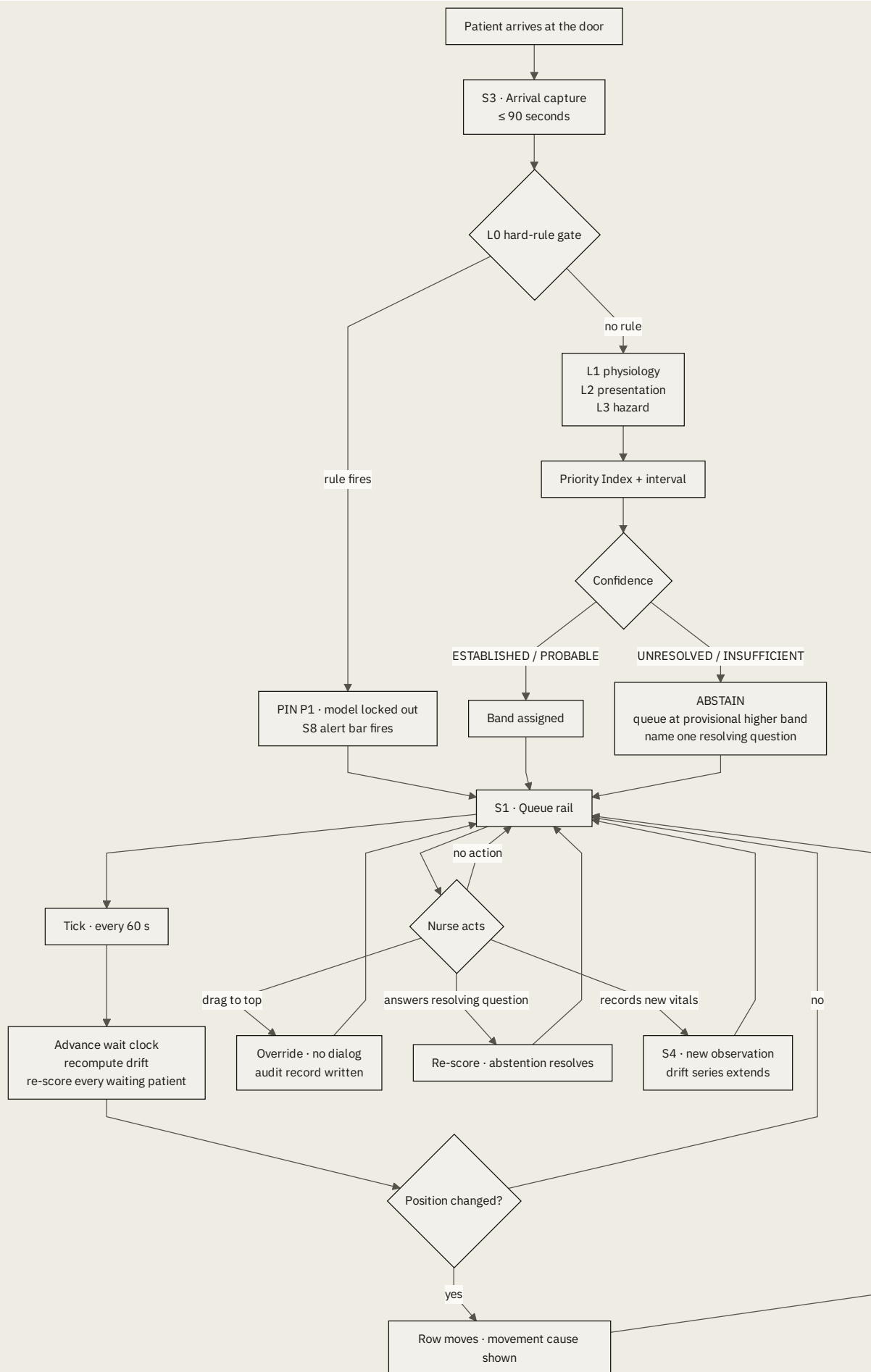
The application is **one page**. There is no router and no navigation. Everything is a region of the board or an overlay on top of it. This is deliberate: a nurse who has to navigate has already lost the two minutes they had.

#	REGION	TYPE	ALWAYS VISIBLE?
S0	Board header	Persistent strip	Yes
S1	Queue rail	Primary region (left, ~62%)	Yes
S2	Inspector	Secondary region (right, ~38%)	Yes — shows the selected row, or the board summary when nothing is selected
S3	Arrival capture	Overlay sheet	On demand
S4	Re-assessment sheet	Overlay sheet	On demand / on prompt
S5	Audit drawer	Bottom drawer	On demand
S6	Fairness monitor	Full-region takeover of S1+S2	On demand
S7	Surge banner	Header-attached strip	Automatic
S8	Absolute-emergency alert	Interruptive bar, top	Automatic, rare
S9	Simulation console	Footer strip	Prototype only

S9 is prototype scaffolding and is visually marked as such. It exists so a judge can drive the demo. It carries a `PROTOTYPE CONTROLS – NOT PART OF THE CLINICAL PRODUCT` label and is the only element in the interface that is allowed to look like a control panel.

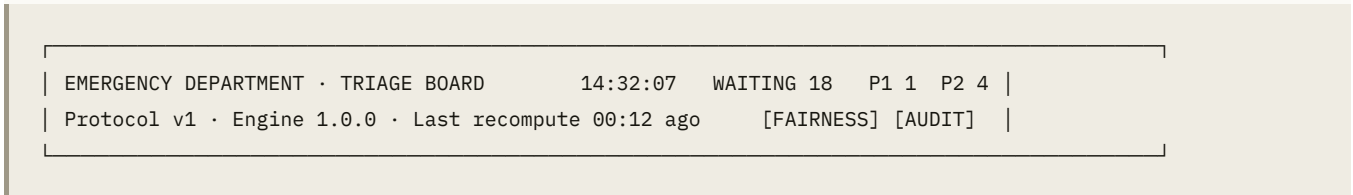
2. Top-level flow





3. S0 · Board header

A single 44 px strip. It is a running header of a clinical document, not a navigation bar.



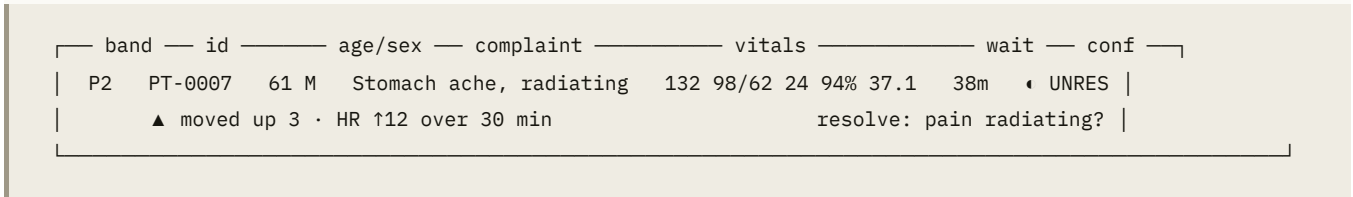
ELEMENT	BEHAVIOUR
Department + board name	Static
Clock	Simulation clock, updates each second, tabular figures so digits do not shift
Census counts	Waiting total, then counts per band, updated on each tick
Protocol / engine version	Always visible. A nurse must be able to say which rule set produced a decision. Tapping opens the protocol file in read-only view.
Last recompute age	Turns to an alarm state if it exceeds 90 s — the interface must never silently stop being live
[FAIRNESS]	Opens S6
[AUDIT]	Opens S5

Trigger: the recompute-age counter is driven by the tick, not by a separate timer. If the tick loop dies, the counter freezes and then alarms. This is intentional — a frozen clock is a visible failure rather than an invisible one.

4. S1 · Queue rail (the primary surface)

One row per waiting patient, ordered by Priority Index descending. This list is the product.

4.1 Row anatomy



ZONE	CONTENT	NOTES
Band chip	P1–P5, or ∅ when abstaining	Never colour alone — the letter+number is the primary carrier
Encounter ID	PT-0007	Monospace, tabular
Age / sex	61 M, 3 y F, 6 mo M, ~40 ? for estimated	Estimated ages carry ~
Complaint	Truncated to one line at the row width	Full text in inspector
Vitals strip	HR · SBP/DBP · RR · SpO ₂ · Temp, monospace, fixed columns	— for unobtainable; each value carries an arrow when it has drifted
Wait	Minutes since arrival	Turns to escalation state when the reassessment interval has elapsed
Confidence	• established · ▶ probable · ◦ unresolved · ◦ insufficient — always with a text token	Two carriers, per NFR-6
Movement line	Appears only when the row moved on the last tick, with the cause	Disappears after 3 ticks
Resolve line	Appears only when abstaining	The single question, verbatim

4.2 Row states

STATE	TRIGGER	VISUAL TREATMENT
Normal	default	Hairline rule below, no fill
Selected	tap / arrow keys	2 px left marker, inspector loads
Moved up	position improved on last tick	Movement line with cause, ▲ marker
Overdue reassessment	wait > interval for band	Wait cell inverts, REASSESS token appears
Abstaining	confidence UNRESOLVED / INSUFFICIENT	Band chip becomes ∅ over the provisional band, resolve line shown
Unresolvable	confidence UNRESOLVABLE	Band chip ∅ over the provisional band; in place of the resolve line, ESCALATE — questioning cannot separate P2 from P3 with the recorded reason. Queued at the provisional band like any abstention; it is never a reason to defer care.
Rule-pinned	any PIN_P1 rule fired	Alarm rule above and below the row, LOCKED token, cannot be sorted below any other row
Overridden	nurse moved it	NURSE token with the nurse's band; engine band shown struck through in the inspector, never hidden
Collapsed	surge mode, rank > 5	Single 20 px line: band, ID, complaint, wait only

4.3 Interaction triggers

GESTURE / KEY	EFFECT
Tap row	Select; inspector loads that encounter
↑ / ↓	Move selection
Long-press + drag, or Shift+↑	Begin override reposition
Drop, or Enter	Commit override. No dialog. No justification field. Audit record written immediately.
Tap [+ ARRIVAL]	Open S3
Tap the vitals strip	Open S4 for that patient
Tap the resolve line	Open a three-button answer for the resolving question (Yes / No / Cannot assess)
Esc	Deselect
A	Open audit drawer
F	Open fairness monitor

Override has no confirmation step by design (PRD FR-4.1). A confirmation dialog is a two-second tax levied on the person the system exists to serve, at exactly the moment they have decided the machine is wrong. The audit log, not the dialog, is what makes the override safe.

5. S2 · Inspector — the derivation panel

This panel is the FDA "independently review the basis" requirement made visible. It reads top to bottom as an argument, not as a dashboard.

PT-0007 · 61 M

Stomach ache, radiating to left arm

Walk-in · arrived 13:54 · waited 38 min

PRIORITY INDEX 63.4

55.1

71.7

CONFIDENCE

• UNRESOLVED – cannot separate P2 from P3

Tie broken upward. Undertriage cost 8:1.

Queued at P2 while unresolved.

ONE QUESTION WOULD RESOLVE THIS

"Does the pain radiate to the jaw, shoulder or arm?"

[YES] [NO] [CANNOT ASSESS]

DERIVATION

L0	Hard rules	none fired	–
L1	Physiology	NEWS2 aggregate 4	+4
	HR 132	score 3	
	SpO2 94%	score 1	
	Temp	NOT OBTAINED – interval widened	
L2	Presentation	abdominal_pain	+6
	Age ≥55 with diaphoresis		+7
	Atypical ACS pattern flagged		
L3	Hazard	38 min at P3 rate 0.10	+3.8
	Drift HR ↑12 over 30 min		+2.1

VITAL TREND (45 min)

HR	120 → 132	↑ rising, 4 of 4 readings
SpO2	96 → 94	↓ falling
RR	20 → 24	↑ rising

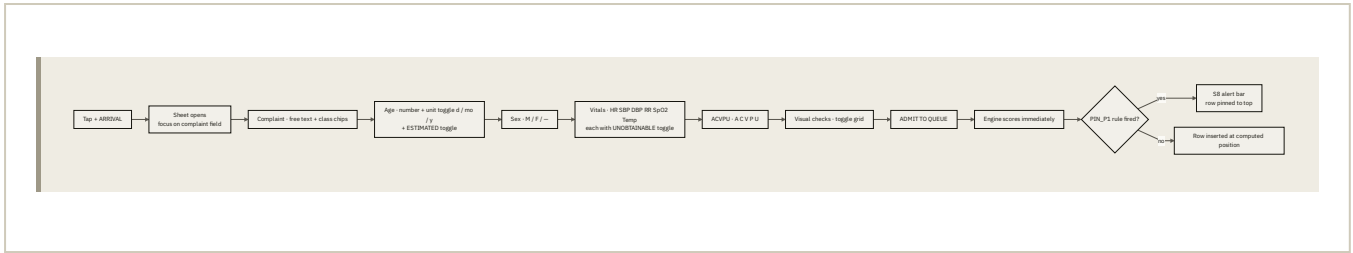
REASSESS DUE 14:47 (in 15 min)

ENGINE 1.0.0 · PROTOCOL v1 · COMPUTED 14:32:07

BEHAVIOUR	RULE
Empty state	When no row is selected, the inspector shows board-level summary: census by band, oldest un-reassessed patient, current mode, override count this shift
Every number is traceable	Each derivation line names its source layer and its point contribution. Nothing appears without a reason.
Missing data is stated, not omitted	NOT OBTAINED – interval widened is a line item, not an absence
Overridden patients	Show both: ENGINE RECOMMENDED P3 (struck) and NURSE ASSIGNED P2, plus the timestamp. The engine's view is never erased.
Answering the resolving question	Immediately re-scores, the abstention resolves or narrows, and the resolution is written to the audit log as an information event

6. S3 · Arrival capture

A single overlay sheet, no steps, no wizard, no pagination. Everything visible at once because a nurse cannot hold a form's state in their head across screens.



Field rules

FIELD	REQUIRED	BEHAVIOUR WHEN ABSENT
Chief complaint	No	Class defaults to <i>unknown</i> , base risk 8. Absence of a complaint is not absence of risk.
Age	No	Falls to <i>~adult</i> with a widened interval and a visible <code>AGE ESTIMATED</code> token
Sex	No	–; sex modifiers do not apply and the interval widens
Each vital	No	Individually markable <code>UNOBTAINABLE</code> ; each one reduces evidence completeness by its weight
ACVPU	Yes	The only mandatory field. It is a look, not a measurement, and it drives two hard rules.
Visual checks	No	Toggles: <i>pale · diaphoretic · drowsy · distressed · cyanosed · active bleeding · cannot speak in full sentences · airway compromise · seizure now · heavy vaginal bleeding (shown only when pregnancy status is pregnant or postpartum)</i>
Complaint qualifiers	No	One-tap chips shown for the selected complaint class: <i>thunderclap · neck stiffness · neurovascular compromise · radiating · exertional · sudden onset · carer reports a change from baseline</i> . These drive presentation modifiers; the engine never parses the free-text complaint.

The zero-history path is the default path. A patient with no name, no age, no history and no ability to speak produces a complete, scoreable record. There is no "unknown patient" special mode because unknown is the normal case.

Capture budget: 90 seconds. Numeric fields use a large keypad, not the OS keyboard. Class chips are one tap. The whole sheet is reachable without scrolling at 1280×800.

7. S4 · Re-assessment

Opens from the vitals strip of a row, or from a reassessment prompt.

Contains only what changes: the five vitals, ACVPU, and the visual grid — pre-filled with the previous values so the nurse edits deltas rather than retyping. Committing appends a new observation to the series, which immediately feeds L3 drift.

Trigger for the prompt: `wait > reassessInterval[band]` . The row enters the *overdue reassessment* state. It does not disappear, and it does not silently lapse — PRD FR-3.6. An un-reassessed patient becomes *more* visible over time, never less.

8. S5 · Audit drawer

Slides up over the lower half. Append-only, newest first, exportable.

COLUMN	CONTENT
Time	Simulation timestamp
Encounter	ID
Event	ARRIVAL · SCORE · RULE_FIRED · OVERRIDE · REASSESS · QUESTION_ANSWERED · MODE_CHANGE · PROTOCOL_CHANGE
Engine said	Band + confidence at that instant
Nurse did	Band assigned, where applicable
Basis	The full derivation snapshot, expandable
Chain	Truncated SHA-256, and the verification state of the chain

Two actions: `EXPORT JSON` and `EXPORT CSV`. Both produce a file a clinician can read without this application (PRD FR-4.4).

Override record shape — the complete system state at the moment of the override, so that a reviewer six months later can reconstruct exactly what the nurse was looking at:

```
{
  "eventId": "EV-000148",
  "type": "OVERRIDE",
  "at": "2026-08-23T14:41:02+05:30",
  "encounterId": "PT-0007",
  "engineBand": "P3",
  "engineConfidence": "UNRESOLVED",
  "engineIndex": 63.4,
  "engineInterval": [55.1, 71.7],
  "rulesFired": [],
  "inputsSnapshot": { "hr": 132, "sbp": 98, "rr": 24, "spo2": 94, "temp": null, "acvpu": "A" },
  "derivationSnapshot": { ... },
  "nurseBand": "P2",
  "nurseId": "STAFF-04",
  "reasonChip": null,
  "prevHash": "9f2c...", "hash": "41ab..."
}
```

`reasonChip` is nullable and is filled after the fact if at all. The override took effect before this record was complete.

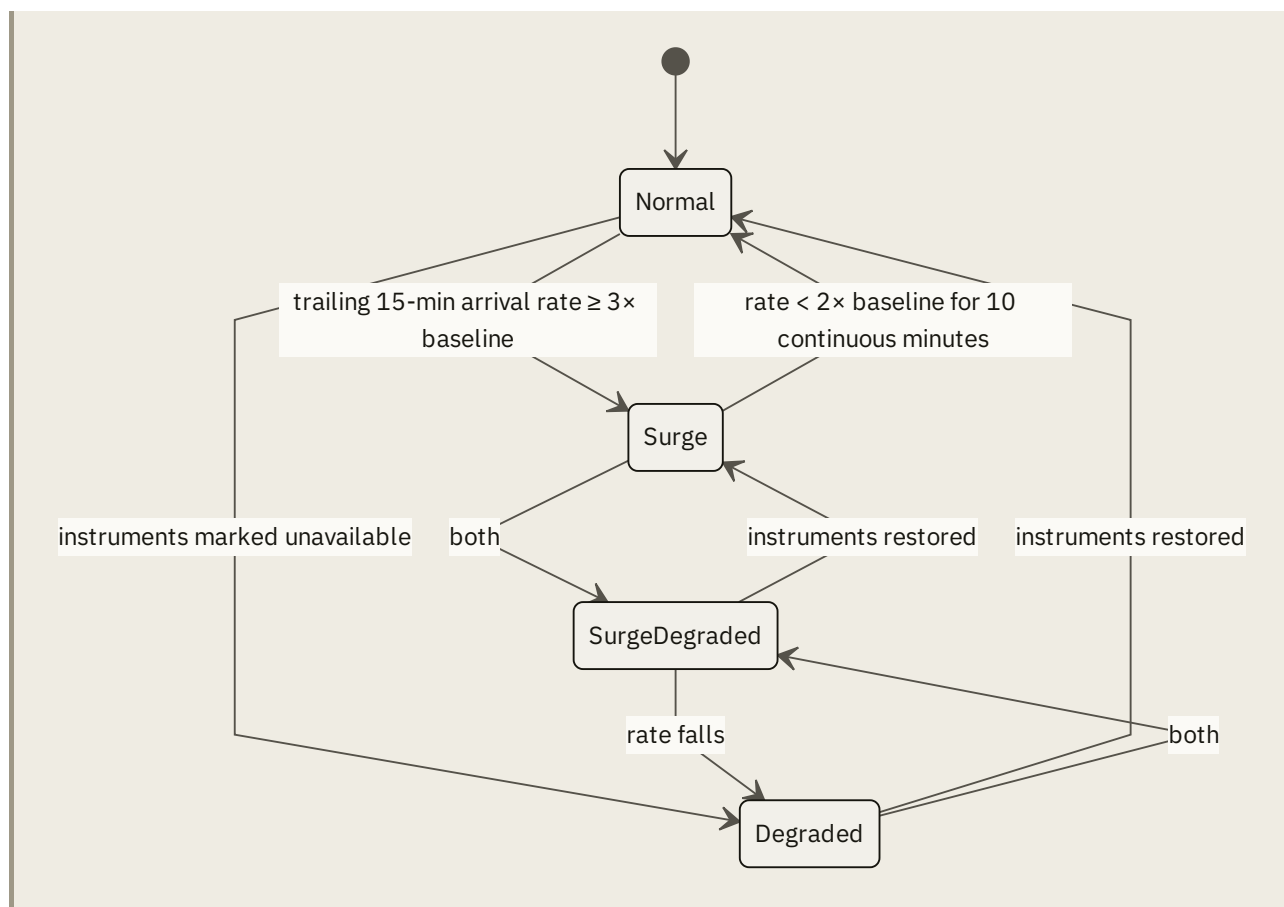
9. S6 · Fairness monitor

Takes over the board region. Three panels, no decorative chrome.

PANEL	CONTENT
Assigned priority by subgroup	Horizontal bars: sex (M/F/—), age band (0–1, 1–5, 5–16, 16–65, 65–80, 80+), language. Distribution of bands per subgroup.
Undertriage proxy	Rate of <i>acuity upgrade after triage</i> per subgroup — the observable stand-in for undertriage. The worst-served subgroup is named in words at the top , not left for the reader to find in a chart.
Drift	Each subgroup's proxy over the shift against its tolerance band. Crossing the tolerance raises a flag with the subgroup named and the encounters reachable.

Wording rule: this panel never reports an average. The headline sentence is always of the form "Women aged 45–65 presenting with chest pain are upgraded after triage at 2.3× the board rate." An aggregate fairness number is how bias hides.

10. S7 · Surge mode



On entering Surge: 1. A header-attached strip states the mode, the trigger and the numbers: `SURGE MODE · 19 arrivals/hour · 3.2× baseline · entered 14:38`. 2. Ranks 1–5 stay in full row form. Everything from rank 6 down collapses to a 20 px line. 3. The inspector switches to a five-patient stack rather than a single-patient detail. 4. Reassessment intervals shorten by 33% for P2 and P3 — crowding is exactly when reassessment lapses, so the system compensates in the opposite direction to the department. 5. A `MODE_CHANGE` audit record is written with the trigger values.

On entering Degraded (instruments unavailable): 1. All instrument-derived vitals render as `—` and score as unobtainable. 2. Evidence completeness falls; intervals widen; more patients enter abstention. 3. The banner names what has been lost: `DEGRADED — no monitors. Scoring from visual assessment and complaint only. Cannot discriminate P2 from P3 for 6 patients.` 4. The queue keeps moving. Degraded is not a stop condition.

11. S8 · Absolute-emergency alert

Fires the instant a `PIN_P1` rule evaluates true, independently of queue position or which screen is open.

- A full-width interruptive bar at the very top, above the header.
- Text: the encounter ID, the rule label, and the measured value that fired it — PT-0013 · RULE-CONSC-01 · ACVPU = U · unresponsive .
- One action: ACKNOWLEDGE . Acknowledgement is logged; it does not clear the pin.
- The row is locked to the top of the queue and **cannot be sorted below any other row**, including by override. A nurse may override a band upward; the interface does not offer a path to move a rule-pinned patient down. (Recorded as an open question in the PRD only insofar as who may edit the rules — not whether the lock holds.)

These alerts are rare by construction. The rules are absolute physiological criteria, not probabilistic thresholds, which is what keeps this the only interruptive element in the interface and keeps alert fatigue out of the design.

12. S9 · Simulation console (prototype only)

A footer strip, visually marked as scaffolding.

CONTROL	EFFECT
▶ /	Run / pause the simulation clock
1× 10× 60×	Clock speed. 60× advances an hour of waiting-room time in a minute — this is how the live re-triage becomes visible in a five-minute pitch.
SURGE ×3	Inject arrivals at 18/hour
LOSE MONITORS	Enter degraded mode
RESET	Reload the cohort at t=0
t + 15 min	Jump the clock forward

13. The demo path (five minutes, in order)

This is the sequence the prototype is built to support, and the order the pitch should follow.

#	ACTION	WHAT IT PROVES
1	Board loads at t=0 with 20 patients	Scoring on a realistic cohort
2	Select PT-0004 (82 F, "just not right")	Geriatric atypical presentation — normal-looking vitals, P2 from presentation risk. Read the derivation aloud.
3	Select PT-0011 (3 y, T 38.5, HR 168)	Age-banded scoring. Show that the same numbers in an adult would score differently.
4	Select PT-0007 (61 M, "stomach ache")	Abstention. The system refuses to guess and names one question. Answer it; watch the band resolve.
5	Run the clock at 60× for ~45 simulated minutes	The queue is alive. PT-0002 rises on RR/SpO ₂ drift with the cause stated on the row.
6	Drag PT-0020 (52 F, chest pain, normal vitals) to the top	The nurse decides. No dialog. Open the audit drawer — the complete captured state is there.
7	Press SURGE ×3	Disaster mode. Top five stay, the rest collapse, the trigger is stated.
8	Press LOSE MONITORS	Degraded mode. Intervals widen, the banner names which bands can no longer be separated, the queue keeps moving.
9	Open the fairness monitor	The worst-served subgroup named in a sentence.
10	Turn off the network and reload	Stage 1 is real. It still works.

Step 10 is the one that ends the argument. Everything before it is a claim about software; step 10 is a claim about a district hospital at 2 a.m., and it is the one that decides whether this deploys.

04

UI / UX Brief

The visual system and the anti-generic contract

Purpose	Every colour, size, weight, spacing value and interaction rule. An implementing agent picks nothing; it looks it up here.
Depends on	01-PRD.md, 03-APP-FLOW.md
Revision	1.0

1. The position

This is a board in an emergency department, read standing, at arm's length, by someone who is being interrupted. It is a clinical document that updates itself. It is not a product page, not a SaaS dashboard, and not a data-visualisation showcase.

Everything below follows from that sentence. When a decision is unclear, the test is: *would this element exist on a well-designed paper form that a hospital had been refining for thirty years?* If not, it does not go on the screen.

1.1 References we are actually designing from

REFERENCE	WHAT WE TAKE
The paper ED triage form and the physical whiteboard	Running header with department, date, shift, revision. Ruled rows. Fixed columns. A colophon at the foot.
Laboratory and test-equipment panels (oscilloscopes, blood-gas analysers, ventilators)	Monospaced tabular numerals as the primary content. Units set smaller than values. Colour reserved for alarm states. Legibility over elegance.
Swiss clinical and transport documentation — Otl Aicher's hospital and airport systems	Hairline rules doing the work that boxes and shadows usually do. Uppercase letterspaced micro-labels. Asymmetric field division.
Airline load sheets, ships' logs, anaesthetic charts	Documents where a number has a consequence, so nothing is decorative and nothing is hidden behind an interaction.

1.2 What we are explicitly not designing from

Stripe's marketing site. Linear's app chrome. Any Tailwind starter. Any dribbble "healthcare dashboard concept". Any interface whose primary visual event is a card with a soft shadow.

2. The anti-generic contract

The current generation of AI-assisted frontends converges on one look, and the tells are well documented ([925 Studios](#), [avoid-ai-design](#), [Superdesign](#)). The convergence has a traceable cause: Tailwind's 2019 `indigo-500` default plus shadcn's untouched `zinc / slate` plus Inter.

This project **must not** produce that. The contract below is binding. A pull request violating any line is rejected.

2.1 Prohibited — no exceptions

#	BANNED	WHY
1	Inter, Roboto, Poppins, Montserrat, Nunito, Space Grotesk, or any system-default-only stack	The single loudest tell. Inter signals no typographic decision was made.
2	Any purple, violet, indigo or #6366F1-adjacent hue, anywhere, at any opacity	The AI-design fingerprint
3	Any gradient. Including <code>bg-clip-text</code> headline gradients and "subtle" background washes.	See above
4	<code>box-shadow</code> of any kind	Elevation here is expressed by a rule and a ground shift, as on paper
5	<code>border-radius</code> above 2 px, and any <code>rounded-2xl</code> equivalent	Sharp corners read as instrument; soft corners read as consumer app
6	Glassmorphism, backdrop-blur, translucent panels	
7	Three feature cards in a row. Any card-grid layout.	This interface has no cards at all
8	Lucide, Feather, Heroicons, Font Awesome, Material Icons	Icon libraries carry a visual fingerprint we are avoiding
9	Emoji, anywhere, including in copy and commit messages	
10	Centred hero layouts, full-width CTA bands, four-column footers	
11	<code>fade-in-up</code> on load, staggered entrance animation, scroll-triggered reveal	Motion here means one thing: a value changed
12	Uniform spacing (<code>gap-4</code> / <code>p-6</code> on everything)	Spatial hierarchy must be deliberate and unequal
13	Stock photography, illustration, gradient placeholders, avatars	
14	Copy in the register of "Build faster. Ship smarter."	Every string is clinical and specific
15	Colour as the sole carrier of any meaning	WCAG 1.4.1, and it is a patient-safety requirement in a department where colour-vision deficiency is unscreened

2.2 The five defining moves

Three to five decisive choices are what separate a designed interface from a defaulted one. Ours:

- Warm paper ground, ink rules, zero shadows.** The background is `#F2F0EA`, not white and not `slate-50`. Structure is carried entirely by 1 px hairlines. There is no card, no panel, no elevated surface anywhere in the application.
- Type is IBM Plex, used as three distinct voices.** Plex Mono for every number and code. Plex Sans Condensed for uppercase letterspaced micro-labels. Plex Serif — used in exactly one place, the inspector's derivation headings — so that the reasoning panel reads as an argument in prose rather than as a widget.
- Colour is a signal, never a surface.** The interface is achromatic. Five hues exist, each bound to one meaning, and each also carries a text token and a glyph. Nothing is coloured because it looks good.
- Asymmetric 62/38 split, and a real table.** Not halves, not thirds, not a card grid. The queue is a genuine fixed-column table so that vitals align vertically down twenty rows — the single property that makes a board scannable and that no card layout can provide.
- The colophon.** A footed line, set in the serif, stating engine version, protocol version, render timestamp and the recommend-not-decide boundary. It is the element that most clearly marks this as a clinical document rather than a web page, and it is load-bearing legally as well as visually.

3. Colour tokens

Defined once at the top of `board.css` as custom properties. **No colour literal appears anywhere else in the stylesheet.**

3.1 Ground and ink — light (canonical)

TOKEN	VALUE	USE
--ground	#F2F0EA	Page. Warm paper, not white.
--ground-raised	#FAF9F5	Overlay sheets, inspector — a shift of ground, never a shadow
--ground-sunk	#E7E4DA	Header strip, footer, table head
--ground-invert	#16150F	Inverted tokens, alarm fills
--ink	#16150F	Primary text
--ink-2	#524F48	Secondary text, units, meta
--ink-3	#706B61	Tertiary, disabled, placeholder
--rule	#CFCABA	Hairlines — the workhorse
--rule-strong	#9E9887	Section divisions, table head underline
--rule-alarm	#A81E12	Rules bracketing a pinned row

3.2 Signal hues — bound to meaning, one each

TOKEN	VALUE	BOUND MEANING	TEXT TOKEN	GLYPH
--sig-alarm	#9B1C11	P1 · rule fired · deterioration	P1 / LOCKED	●
--sig-escalate	#A3590A	P2 · overdue reassessment	P2 / REASSESS	▲
--sig-steady	#524F48	P3 — deliberately achromatic	P3	—
--sig-low	#706B61	P4, P5	P4 P5	○
--sig-abstain	#1F5673	Uncertainty, abstention, resolving question	UNRESOLVED	⊗
--sig-nurse	#2E5A3E	Human override, nurse-assigned band	NURSE	‡

Two additional tokens exist for the single filled element in the system, the P1 band chip (§6.1). They are separate tokens because a hue that is legible *as text on paper* and a hue that is legible *as a background under paper-coloured text* are not the same hue, and conflating them is how filled alarm states end up unreadable.

TOKEN	LIGHT	DARK	USE
--sig-alarm-fill	#9B1C11	#8E1A10	P1 chip background
--sig-alarm-fill-ink	#F2F0EA	#EDEAE0	Text on the P1 chip

P3 has no colour. Most triage boards give the middle band a yellow, which trains the eye to see three-quarters of the list as "warning" and destroys the alarm signal. Here the middle of the distribution is ink, and colour only appears where something is actually wrong or actually uncertain. This is the choice that makes the alarm states work.

3.3 Dark variant

Available via `prefers-color-scheme: dark` for night shift. Light is canonical; dark is a faithful inversion, not a redesign. Signal hues are lifted for contrast on dark ground and their meanings are unchanged.

TOKEN	DARK VALUE
--ground	#14130E
--ground-raised	#1C1B15
--ground-sunk	#0E0D09
--ink	#EDEAE0
--ink-2	#A5A093
--ink-3	#837F71
--rule	#33312A
--rule-strong	#4E4B41
--sig-alarm	#EF6F62
--sig-escalate	#DE9333
--sig-steady	#A5A093
--sig-low	#837F71
--sig-abstain	#5C9CBF
--sig-nurse	#6FA57F

3.4 Contrast requirements — measured, not asserted

Every value in §3.1–§3.3 has been computed against the WCAG 2.1 relative-luminance formula and passes. **These are measurements, not intentions.** Any token change must be re-measured before merge, and the measurement belongs in the pull request.

PAIR	REQUIRED	LIGHT	DARK
--ink on --ground	≥ 12:1	16.05	15.45
--ink-2 on --ground	≥ 7:1	7.17	7.13
--ink-3 on --ground	≥ 4.5:1	4.65	4.64
--sig-alarm on --ground	≥ 4.5:1	7.18	6.31
--sig-escalate on --ground	≥ 4.5:1	4.62	7.37
--sig-steady on --ground	≥ 4.5:1	7.17	7.13
--sig-low on --ground	≥ 4.5:1	4.65	4.64
--sig-abstain on --ground	≥ 4.5:1	6.99	6.17
--sig-nurse on --ground	≥ 4.5:1	6.96	6.52
--sig-alarm-fill-ink on --sig-alarm-fill	≥ 7:1	7.18	7.57

Why 4.5:1 for signal hues and 7:1 only for the P1 fill. An earlier revision of this brief demanded 7:1 of every signal hue. That requirement is not reachable while the hues remain what they are: amber at 7:1 on warm paper is `#784107`, which reads as brown and stops working as an escalation signal, and a dark-mode red at 7:1 on near-black is `#EA847B`, a washed

salmon that is a worse alarm than a saturated red. Demanding a ratio that forces a colour to stop meaning what it means is a specification error, not a safety measure.

The requirement that carries the actual safety weight is on the **one interruptive element** — the filled P1 chip, where paper-coloured text sits on a red ground — and that pair is held to 7:1 in both modes. Everything else is held to WCAG AA for normal text (4.5:1), which is AAA for the ≥ 18.66 px sizes most of these hues are used at, and is backed by the rule that colour is never the sole carrier of meaning (§3.2, NFR-6).

Verify with: `L = 0.2126R + 0.7152G + 0.0722B` on linearised sRGB channels; ratio $(L_1+0.05)/(L_2+0.05)$.

4. Typography

4.1 Families

```
--font-data: "IBM Plex Mono", "SFMono-Regular", "Menlo", "Consolas", monospace;
--font-ui:    "IBM Plex Sans", "Helvetica Neue", "Segoe UI", system-ui, sans-serif;
--font-label: "IBM Plex Sans Condensed", "IBM Plex Sans", "Helvetica Neue", sans-serif;
--font-prose: "IBM Plex Serif", "Iowan Old Style", "Georgia", serif;
```

IBM Plex is chosen, not defaulted: it was drawn for an engineering organisation, its mono is a genuine text-grade monospace rather than a code face, and its condensed cut gives us a label voice most UI families do not have. It is also not on anyone's list of AI-default typefaces.

Loaded from Google Fonts and precached by the service worker so the offline claim holds. The fallback stack has matched metrics; layout does not shift when the webfont is absent.

4.2 Scale — chosen sizes, not a ratio applied mechanically

TOKEN	SIZE / LINE-HEIGHT	FAMILY	WEIGHT	TRACKING	USE
--t-micro	10 / 12	label	600	+0.14em, uppercase	Field labels, column heads, tokens
--t-meta	11 / 15	ui	400	+0.01em	Timestamps, versions, footnotes
--t-body	13 / 19	ui	400	0	Interface prose
--t-row	15 / 18	ui	400	0	Row complaint text
--t-data	17 / 18	data	500	0	Vitals. The most important size in the interface.
--t-band	22 / 22	data	600	+0.02em	Band chip
--t-index	30 / 30	data	400	-0.01em	Priority Index in the inspector
--t-head	14 / 20	prose	600	0	Inspector derivation headings — the only serif

4.3 Numeral rules

All numeric content uses `font-variant-numeric: tabular-nums lining-nums;` . Non-negotiable: a clock or a heart rate whose digits change width makes the eye re-find the value on every tick, and on a board that ticks every second that is a genuine legibility failure.

Units are set at `--t-meta` in `--ink-3` , immediately after the value with no space: `132bpm` renders as `132` + a small `bpm` . The value dominates; the unit is available but never competes.

Unobtainable values render as `--` (two em-dashes) in `--ink-3`, never as `0`, `N/A`, `null`, or an empty cell. An empty cell is indistinguishable from a rendering failure.

4.4 Copy register

Terse, clinical, declarative. Sentence case for prose; uppercase only for micro-labels and tokens.

WRITE	NEVER WRITE
"Not obtained. Interval widened."	"No data available 🙄"
"Cannot discriminate P2 from P3."	"Low confidence"
"Tie broken upward. Undertriage cost 8:1."	"Erring on the side of caution"
"Reassess due 14:47."	"Don't forget to check on this patient!"
"Queued at P2 while unresolved."	"Provisional priority assigned"
"One question would resolve this."	"Need more info?"

Every string states a fact and, where relevant, the consequence. No string apologises, encourages, or celebrates.

5. Space and structure

5.1 Spacing scale — deliberately unequal

```
--s1: 3px;  --s2: 6px;  --s3: 9px;  --s4: 14px;
--s5: 20px; --s6: 30px;  --s7: 48px;
```

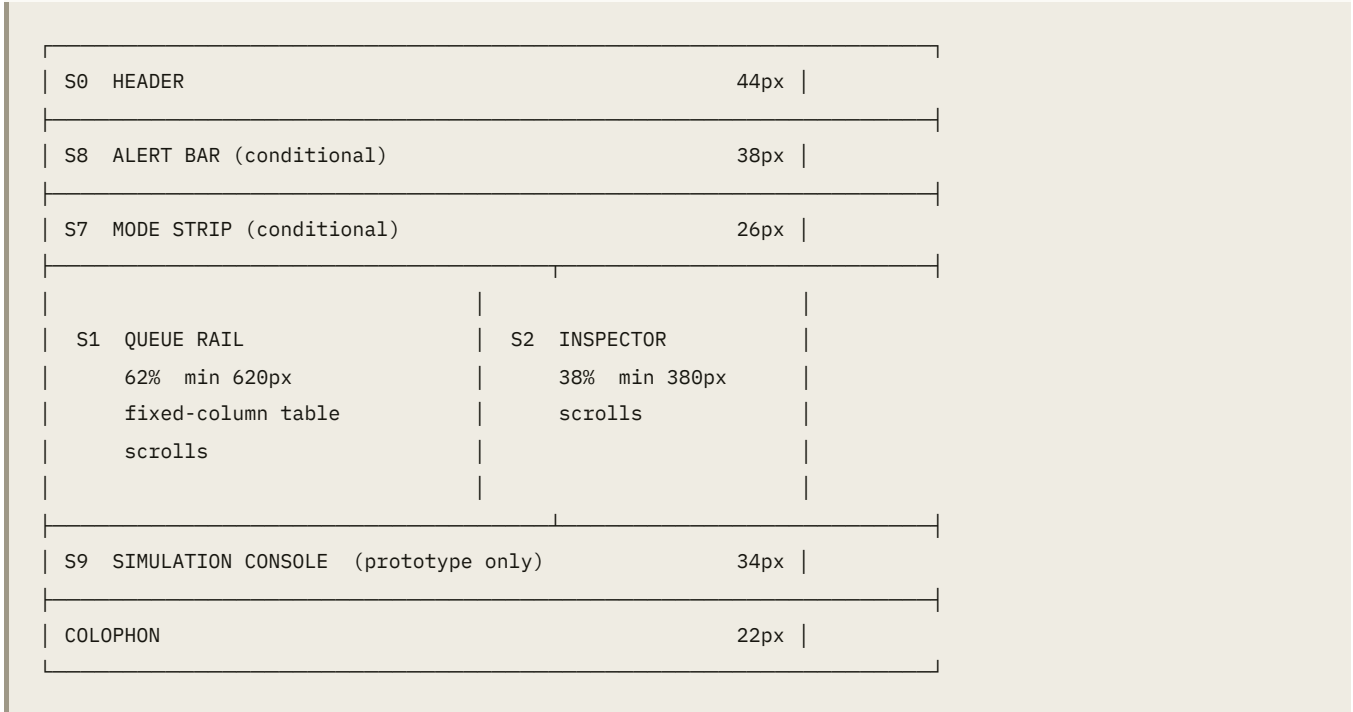
A 3 px base with non-linear steps. Uniform `gap-4` everywhere is an explicit tell (§2.1 #12); hierarchy must be visible in the spacing itself. Row internal padding is `--s2` vertical, `--s3` horizontal. Region padding is `--s5`. The gap between the queue and the inspector is a 1 px rule plus `--s5` on each side — not a gutter of empty space.

5.2 Radius and elevation

```
--radius: 0;          /* everything */
--radius-control: 2px; /* only: inputs, chips, buttons */
```

There is no `--shadow` token. Shadows do not exist in this system. A surface that needs to read as raised gets `--ground-raised` and a 1 px `--rule-strong` border. This one rule removes the largest single component of the generic look.

5.3 Layout



`grid-template-columns: minmax(620px, 62fr) 1px minmax(380px, 38fr);` — the 1 px middle track is the divider rule itself, not a border on a neighbour, so it stays exactly 1 device pixel at every zoom level.

Below 1024 px the inspector becomes a bottom drawer over the queue. The queue never collapses; it is the product.

5.4 The queue table

A real `<table>` with `table-layout: fixed`. Semantics matter for screen readers, and fixed columns are what make twenty rows of vitals scannable.

COLUMN	WIDTH	ALIGN
Band	54 px	centre
ID	84 px	left
Age/Sex	62 px	left
Complaint	1fr	left
HR	58 px	right
BP	78 px	right
RR	48 px	right
SpO ₂	58 px	right
Temp	58 px	right
Wait	56 px	right
Conf	96 px	left

Row separation is a 1 px `--rule` under each row. **No zebra striping** — alternating background fills compete with the alarm states, which are also background changes. On a board where a fill means "this patient is critical", decorative fills are a safety problem.

Row height 42 px normal, 20 px collapsed in surge mode. Touch target for the whole row ≥ 44 px including padding.

6. Component specifications

6.1 Band chip

54 × 26 px, `--radius-control`, `--t-band`, mono 600. Normal bands: 1 px border in the signal hue, transparent ground, hue-coloured text. **P1 only:** filled `--sig-alarm-fill`, text `--sig-alarm-fill-ink`. Exactly one band in the system is a filled block, and it is the one that means *now*.

Abstaining: the chip shows `∅` in `--sig-abstain` with the provisional band beneath it at `--t-micro`.

6.2 Confidence indicator

Glyph plus text token, always both. Five states:

DISPLAY	HUE	MEANING
● ESTABLISHED	<code>--ink</code>	Interval sits inside one band
▸ PROBABLE	<code>--ink</code>	Crosses one boundary, ≥ 65:35
◐ UNRESOLVED	<code>--sig-abstain</code>	Abstains. A question is offered.
◐ UNRESOLVABLE	<code>--sig-escalate</code>	Abstains. Questioning cannot help — escalate.
○ INSUFFICIENT	<code>--sig-abstain</code>	Too little was obtained to discriminate

Set at `--t-micro`. Never a percentage — false precision on an uncertainty measure is worse than no measure.

`UNRESOLVED` and `UNRESOLVABLE` share the ◐ glyph deliberately: the glyph encodes *how uncertain*, and these two are equally uncertain. What separates them is what the nurse should do, so the difference is carried by the text token and the hue — amber, because unlike every other abstention state this one asks for an action the nurse cannot take alone. An `UNRESOLVABLE` row also carries an `ESCALATE` token beside the wait cell.

6.3 Confidence band (inspector)

A 3 px horizontal rule spanning the interval on a 0–100 track, with band boundaries marked as 1 px ticks and the point estimate as a 7 px filled square. Drawn as inline SVG. It is a *band*, physically, because that is what it represents — not a badge, not a percentage, not a progress bar.

6.4 Vital cell

Value in `--t-data`, unit in `--t-meta` `--ink-3`. When drifted since the previous observation, a 9 px ▲ / ▼ in `--sig-alarm` / `--sig-abstain` precedes the value, and the value carries a 140 ms flash on change (§7).

6.5 Sparkline

28 × 14 px inline SVG, 1 px stroke `--ink-2`, no fill, no axis, no grid, no dots except a 2 px terminal marker. Six to eight points. It exists to answer "which way" in a glance, not to be read quantitatively — the numbers beside it do that.

6.6 Fairness bars

Horizontal, 1 px `--rule-strong` baseline, bars in `--ink-2` at 10 px height, the flagged worst-served subgroup in `--sig-escalate`. Values printed at the bar end in mono. No axis lines, no legend, no gridlines — the label is on the bar.

6.7 Buttons

Text set in `--t-micro` uppercase, 1 px `--rule-strong` border, `--radius-control`, `--ground` fill, `--s2 / --s4` padding. Hover: border to `--ink`. Active: ground to `--ground-sunk`. Focus: 2 px `--sig-abstain` outline, offset 1 px.

There is exactly **one** filled button in the application: `ADMIT TO QUEUE` in the arrival sheet, filled `--ink`. One primary action per screen, and it is the one that puts a person in the queue.

6.8 Overlay sheets

`--ground-raised`, 1 px `--rule-strong` border, no shadow, no backdrop blur. The page behind dims via a `--ground-invert` overlay at 22% opacity. No entrance animation — the sheet is present or it is not.

7. Motion

Two animations exist. There are no others.

#	ANIMATION	DURATION	EASING	TRIGGER
1	Value flash — a changed number's background lifts to a 14% tint of its drift hue and decays	140 ms in, 500 ms out	<code>linear</code> out	A vital, index or band value changes on a tick
2	Row translate — a row that changed rank animates from its old Y to its new one	180 ms	<code>cubic-bezier(.2,0,.2,1)</code>	Rank change on a tick

Both exist for one reason: on a board that recomputes every sixty seconds, a value that changes without a visible transition is a change the nurse did not see. Motion here is a safety feature with a specific job.

No fade-in on load. No stagger. No scroll reveal. No hover lift. No skeleton shimmer. `prefers-reduced-motion: reduce` disables both and substitutes a 1 px left marker that persists for three ticks.

8. Iconography

Five typographic marks, no icon library, no emoji:

MARK	MEANING
▲ ▼	Drift direction, rank movement
● ◐ ◑ ◒	Confidence, four levels
⊖	Abstention
‡	Nurse override
—	Not obtained

Two hand-drawn inline SVGs, 1 px stroke, 14 × 14, `currentColor`: export, and close. That is the complete icon inventory.

9. Accessibility

REQUIREMENT	IMPLEMENTATION
Colour never sole carrier	Every state has hue + text token + glyph (§3.2)
Contrast	§3.4, verified with axe
Keyboard	Full operation: ↑/↓ select, Shift+↑/Shift+↓ reposition, Enter commit override, A audit, F fairness, Esc dismiss. Visible 2 px focus ring on every interactive element.
Screen reader	The queue is a real table with <caption>, scope="col" headers, and aria-rowindex. Rank changes announce via aria-live="polite"; a fired hard rule announces via aria-live="assertive".
Touch targets	≥ 44 × 44 px
Text scaling	Layout holds to 200% browser zoom
Reduced motion	§7
Colour-vision deficiency	Alarm red and abstain blue are separable under deuteranopia and protanopia; the achromatic P3 means most of the board is unaffected by any CVD

10. The colophon

Fixed at the foot of the page, `--t-meta` in `--font-prose`, `--ink-3`, on `--ground-sunk`:

PatientTriage.ai · engine 1.0.0 · protocol v1 · board rendered 14:32:07 IST · This display is decision support. It does not assign a triage category, move a patient, or order treatment. The attending clinician's judgement governs.

It states the engine and protocol versions a reviewer would need, timestamps the render, and carries the recommend-not-decide boundary that keeps the product on the non-device side of the FDA CDS line. It is also, incidentally, the single element most responsible for the interface reading as a piece of clinical software rather than a web page.

11. Review checklist

Before any merge:

- [] No purple, indigo, violet or gradient anywhere in the diff
- [] No `box-shadow`; no `border-radius` above 2 px
- [] No Inter / Roboto / Poppins / Space Grotesk
- [] No icon library import; no emoji in code, copy or commit message
- [] No card component; no three-across grid
- [] Every new colour is an existing token — no literals outside the token block
- [] Every new state has hue **and** text token **and** glyph
- [] Every numeric element uses `tabular-nums`
- [] No new animation beyond the two in §7
- [] Every new string passes the copy-register table in §4.4
- [] Contrast verified on both light and dark
- [] Full keyboard path still works

Design-tell sources: [925 Studios — AI slop design tells](#) · [avoid-ai-design \(GitHub\)](#) · [Superdesign — why AI design looks generic](#) · [Why every AI-built website looks the same](#) · Clinical UI: [Medical device UX design guide](#)

05

Backend Schema

Models, relations, authorisation, retention

Purpose	The complete data contract. Stage 1 is device-local and has no server; Stage 2 introduces one. Both are specified here so that Stage 1 is not built into a corner.
Depends on	01-PRD.md, 02-TRD.md
Revision	1.0

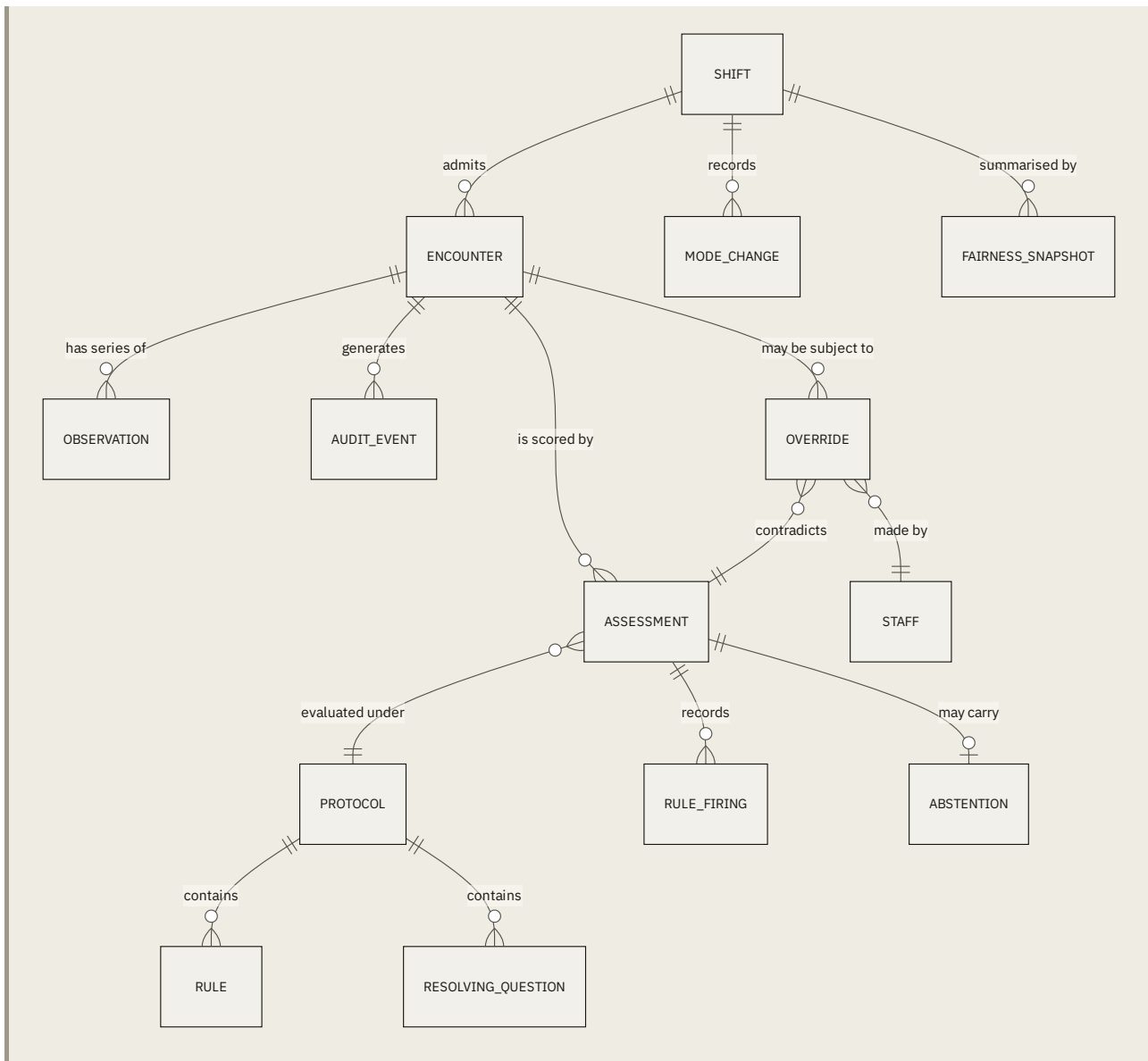
1. Stage-1 principle: the schema is local, and it is minimal

Stage 1 has **no backend**. Everything lives on the tablet. This is not a limitation to be apologised for — it is the deployment story (PRD §10, "one tablet, offline") and it is also the strongest privacy posture available: data that never leaves the device cannot be breached in transit or at rest on someone else's infrastructure.

STORE	MECHANISM	CONTENTS	LIFETIME
session	localStorage	Current shift, board mode, UI preferences	Until shift close
encounters	localStorage (JSON)	Live waiting-room encounters	24 h, then purged
observations	localStorage (JSON)	Vital-sign series per encounter	24 h
audit	IndexedDB, object store audit	Append-only, hash-chained event log	7 years, de-identified at shift close
protocol	Static JSON, fetched	Hospital-owned rules and thresholds	Versioned artefact
cohort	Static JSON, fetched	Prototype only — synthetic encounters	Prototype only

Everything in this document is written so that the same entities move to PostgreSQL at Stage 2 without a field changing name or meaning.

2. Entity relationship model



The relation that carries the product: one encounter has *many* assessments. A triage category in a conventional system is a single value on a patient row. Here it is a time series, because the central claim is that a frozen category is the hazard.

ENCOUNTER has no `band` column — the current band is always `latest(ASSESSMENT).band`. Making that structurally impossible to shortcut is a design decision, not an accident.

3. Core entities

3.1 ENCOUNTER

One person, one visit. **No identity is required.**

FIELD	TYPE	NULL	NOTES
encounter_id	TEXT PK	no	PT-0007. Locally generated, sequential per shift, meaningless off-device.
shift_id	TEXT FK	no	
arrived_at	TIMESTAMPZ	no	
arrival_mode	ENUM	yes	walk_in ambulance referred police unknown
age_value	INTEGER	yes	Null permitted — unidentified patients are the normal case
age_unit	ENUM	yes	days months years
age_estimated	BOOLEAN	no	Default false. Drives the ~ prefix and a widened interval.
age_band	ENUM	no	Derived, never entered: neonate infant toddler preschool school adolescent adult older_adult elderly. Computed before any scoring table is selected.
sex	ENUM	yes	M F X unknown
pregnancy_status	ENUM	yes	not_pregnant pregnant postpartum unknown
gestation_weeks	INTEGER	yes	
complaint_text	TEXT	yes	Free text as spoken. Retained verbatim — the phrasing is clinically informative.
complaint_class	ENUM	no	Default unknown, base risk 8. Absence of a complaint is not absence of risk.
language	TEXT	yes	BCP-47. Collected solely to power the fairness monitor.
preexisting_flags	TEXT[]	yes	diabetic anticoagulated immunosuppressed cardiac_history. Only when volunteered — never required.
complaint_qualifiers	TEXT[]	yes	One-tap chips from arrival capture: thunderclap neck_stiffness neurovascular_compromise radiating exertional sudden_onset reported_change_from_baseline. Presentation modifiers read this; the engine never parses complaint_text.
disposition	ENUM	yes	waiting_in_treatment discharged left_without_being_seen transferred
disposition_at	TIMESTAMPZ	yes	
created_at	TIMESTAMPZ	no	

Deliberately absent: name, MRN, national ID, address, phone, next of kin, insurance, photograph. None is needed to triage, so none is collected. This is data minimisation as architecture rather than as policy.

`age_band` being a stored derived column matters: a fever of 38.5 °C carries different urgency in a 3-year-old than in a 75-year-old, and every scoring path must branch on the band *before* it reaches a threshold table. Storing it makes the branch auditable after the fact.

3.2 OBSERVATION

One set of vitals at one moment. An encounter accumulates a series; the series is what produces drift.

FIELD	TYPE	NULL	NOTES
observation_id	TEXT PK	no	
encounter_id	TEXT FK	no	
observed_at	TIMESTAMPTZ	no	
sequence	INTEGER	no	0 = arrival set
hr	INTEGER	yes	bpm
sbp	INTEGER	yes	mmHg
dbp	INTEGER	yes	mmHg
rr	INTEGER	yes	/min
spo2	INTEGER	yes	%
spo2_on_oxygen	BOOLEAN	yes	NEWS2 scores supplemental oxygen separately
temp_c	NUMERIC(3,1)	yes	°C
acvpu	ENUM	no	A C V P U. The only mandatory clinical field — it is a look, not a measurement, and it drives two hard rules.
pain_score	INTEGER	yes	0–10
cap_refill_s	NUMERIC(2,1)	yes	Paediatric perfusion
unobtainable	TEXT[]	no	Which fields were attempted and could not be obtained
visual	JSONB	no	Toggle set, below
source	ENUM	no	nurse_manual monitor_feed simulated
recorded_by	TEXT FK	yes	STAFF-04

`unobtainable` is a distinct concept from `NULL`. `NULL` means not recorded; `unobtainable` means attempted and failed — no working monitor, patient combative, cuff size unavailable. Only `unobtainable` triggers degraded-mode handling and the widened interval. Conflating the two would let a skipped field silently masquerade as a device failure.

`visual` shape:

```
{
  "pale": false, "diaphoretic": true, "drowsy": false, "distressed": true,
  "cyanosed": false, "active_major_bleeding": false,
  "cannot_speak_full_sentences": false, "airway_compromise": false,
  "seizure_active": false, "heavy_vaginal_bleeding": false
}
```

3.3 ASSESSMENT

One engine run. **Immutable.** Never updated — a new run creates a new row. This is what makes the priority history reconstructible.

FIELD	TYPE	NULL	NOTES
assessment_id	TEXT PK	no	
encounter_id	TEXT FK	no	
observation_id	TEXT FK	no	Latest observation used
computed_at	TIMESTAMPTZ	no	
engine_version	TEXT	no	Semver. Recorded on every score (PRD NFR-8).
protocol_version	TEXT FK	no	
priority_index	NUMERIC(5,2)	no	0–100
interval_low	NUMERIC(5,2)	no	
interval_high	NUMERIC(5,2)	no	
band	ENUM	yes	P1–P5. Null when abstaining — and that is the point.
provisional_band	ENUM	no	Always populated. Drives queue position even under abstention.
band_set_by	ENUM	no	model · hard_rule · presentation_floor · nurse_override. Which mechanism decided the band. "P2 because a rule fired" and "P2 because the index landed there" are different claims, and a reviewer needs to know which one they are reading.
candidate_bands	TEXT[]	yes	Populated when abstaining
confidence	ENUM	no	ESTABLISHED PROBABLE UNRESOLVED UNRESOLVABLE INSUFFICIENT
no_question_reason	ENUM	yes	all_questions_already_answered · no_question_above_information_threshold · no_questions_defined_for_class. Required when and only when confidence = 'UNRESOLVABLE'.
expected_information_gain	NUMERIC(3,2)	yes	0–1, for the question that was offered
evidence_completeness	NUMERIC(3,2)	no	0–1
model_locked_out	BOOLEAN	no	True when a PIN_P1 rule fired
tie_broken_upward	BOOLEAN	no	
layer1_physiology	NUMERIC(4,2)	no	
layer2_presentation	NUMERIC(4,2)	no	
layer3_hazard	NUMERIC(4,2)	no	
derivation	JSONB	no	Full human-readable derivation
resolving_question_id	TEXT FK	yes	
reassess_due_at	TIMESTAMPTZ	no	

Database-level invariants — these enforce PRD principles P1, P3 and P5 in the schema itself, so that no application bug can produce a record that violates them:

```

CONSTRAINT chk_confidence_present
    CHECK (confidence IS NOT NULL),

CONSTRAINT chk_band_or_provisional
    CHECK (band IS NOT NULL OR provisional_band IS NOT NULL),

CONSTRAINT chk_abstention_names_candidates
    CHECK (confidence NOT IN ('UNRESOLVED','UNRESOLVABLE','INSUFFICIENT')
           OR (band IS NULL AND array_length(candidate_bands,1) >= 2)),

-- UNRESOLVED always names a question. This constraint is not weakened;
-- the case with no question is a different state, below.
CONSTRAINT chk_unresolved_names_question
    CHECK (confidence <> 'UNRESOLVED' OR resolving_question_id IS NOT NULL),

CONSTRAINT chk_unresolvable_names_reason
    CHECK (confidence <> 'UNRESOLVABLE'
           OR (resolving_question_id IS NULL AND no_question_reason IS NOT NULL)),

CONSTRAINT chk_reason_only_when_unresolvable
    CHECK (no_question_reason IS NULL OR confidence = 'UNRESOLVABLE'),

CONSTRAINT chk_locked_out_is_p1
    CHECK (NOT model_locked_out OR provisional_band = 'P1'),

CONSTRAINT chk_interval_contains_point
    CHECK (priority_index BETWEEN interval_low AND interval_high)

```

The last one has caught more engine bugs in development than any test.

3.4 RULE_FIRING

Which hard rules fired, on which assessment, on what measured value.

FIELD	TYPE	NOTES
firing_id	TEXT PK	
assessment_id	TEXT FK	
rule_id	TEXT FK	RULE-CONSC-01
protocol_version	TEXT FK	
matched_field	TEXT	acvpu
matched_value	TEXT	U
action	ENUM	PIN_P1 FLOOR_P2 FLAG
alerted_at	TIMESTAMPTZ	
acknowledged_by	TEXT FK, null	Acknowledgement is logged; it does not clear the pin
acknowledged_at	TIMESTAMPTZ, null	

3.5 OVERRIDE

The nurse's decision. The most important table in the system — it is the evidence that the human, not the machine, holds authority.

FIELD	TYPE	NULL	NOTES
override_id	TEXT PK	no	
encounter_id	TEXT FK	no	
assessment_id	TEXT FK	no	The exact assessment being contradicted
staff_id	TEXT FK	no	
at	TIMESTAMP TZ	no	
engine_band	ENUM	yes	Copied, not joined — the record must be readable standalone
engine_confidence	ENUM	no	Copied
engine_index	NUMERIC(5,2)	no	Copied
nurse_band	ENUM	no	
direction	ENUM	no	Derived: upgrade downgrade lateral
reason_chip	TEXT	yes	Nullable by design. Attached afterwards if at all. The move took effect before this field existed.
inputs_snapshot	JSONB	no	Full state the nurse was looking at
derivation_snapshot	JSONB	no	

`reason_chip` being nullable is a product decision expressed as a schema decision. A `NOT NULL` here would make a justification field mandatory, which would put a dialog in front of the override, which would break PRD FR-4.1. The schema is where that promise is kept or broken.

`direction` is what makes the calibration loop work: a systematic upgrade pattern for a given complaint class and subgroup is the signal that the model is undertriaging that group, and it is fed back into threshold review — the model learns from the nurse, never the reverse.

3.6 PROTOCOL , RULE , RESOLVING_QUESTION

The hospital-owned artefact. **Not code.** A deploying hospital changes its red-flag thresholds by editing a versioned JSON file that goes through its own clinical governance, without a software release.

```

{
  "protocolVersion": "v1",
  "hospital": "REFERENCE-ED",
  "effectiveFrom": "2026-08-01",
  "approvedBy": "ED Clinical Governance Committee",
  "supersedes": null,
  "populations": ["adult", "paediatric", "obstetric"],
  "rules": [
    {
      "id": "RULE-CONSC-01",
      "label": "Unresponsive or pain-responsive",
      "population": "all",
      "condition": { "field": "acvpu", "op": "in", "value": ["P", "U"] },
      "action": "PIN_P1",
      "alert": "immediate",
      "rationale": "ESI v5 decision point A – requires immediate lifesaving intervention.",
      "source": "ESI Handbook v5 2023; local protocol v1"
    }
  ],
  "bandThresholds": { "P1": 82, "P2": 62, "P3": 38, "P4": 18 },
  "hazardRates": { "P2": 0.22, "P3": 0.10, "P4": 0.035, "P5": 0.012 },
  "reassessMinutes": { "P2": 15, "P3": 30, "P4": 60, "P5": 120 },
  "costMatrix": { "undertriage": 8, "overtriage": 1 },
  "surge": { "baselineArrivalsPerHour": 6, "multiplier": 3,
    "reassessCompressionPct": 33 },
  "resolvingQuestions": [
    {
      "id": "RQ-ABDO-01",
      "complaintClass": "abdominal_pain",
      "question": "Does the pain radiate to the jaw, shoulder or arm?",
      "discriminatesBetween": ["P2", "P3"],
      "expectedShiftIfYes": 14,
      "expectedShiftIfNo": -6
    }
  ]
}

```

Every threshold that a clinician might reasonably disagree with lives in this file. Nothing clinically contestable is hard-coded. `protocolVersion` appears in the board header at all times and on every assessment, so a reviewer can always answer "which rule set produced this decision".

3.7 AUDIT_EVENT

Append-only, hash-chained, tamper-evident.

FIELD	TYPE	NOTES
event_id	TEXT PK	EV-000148, monotonic
sequence	INTEGER	Gapless, per shift
shift_id	TEXT FK	
at	TIMESTAMP TZ	
type	ENUM	ARRIVAL SCORE RULE_FIRED ALERT_ACK OVERRIDE REASSESS QUESTION_ANSWERED MODE_CHANGE PROTOCOL_CHANGE DISPOSITION EXPORT
encounter_id	TEXT FK, null	Null for shift-level events
actor	TEXT	STAFF-04 or SYSTEM
payload	JSONB	Complete event body
prev_hash	TEXT	SHA-256 of the previous record
hash	TEXT	SHA-256(sequence at type canonical_json(payload) prev_hash)

Chaining. Each record's hash covers its predecessor's. Altering or removing any record breaks every subsequent hash, and the chain is verifiable offline with a fifteen-line script. Computed with `crypto.subtle.digest` on-device; no server is involved and no key management is required, because the property we need is *detection*, not prevention.

`UPDATE` and `DELETE` are not permitted on this table under any role. At Stage 2 this is enforced by grant, not by convention (§5).

3.8 SHIFT , STAFF , MODE_CHANGE , FAIRNESS_SNAPSHOT

ENTITY	KEY FIELDS
SHIFT	shift_id, department_id, started_at, ended_at, protocol_version, baseline_arrivals_per_hour
STAFF	staff_id (pseudonymous), role (triage_nurse charge_nurse physician administrator auditor), department_id. No names. A pseudonymous ID is sufficient for accountability and avoids building a staff-surveillance dataset.
MODE_CHANGE	shift_id, at, from_mode, to_mode, trigger (JSONB: the actual measured values), auto
FAIRNESS_SNAPSHOT	shift_id, at, dimension (sex age_band language), subgroup, n, mean_assigned_band, upgrade_after_triage_rate, tolerance, flagged, is_worst_served

`FAIRNESS_SNAPSHOT` stores per-subgroup rows and never a board aggregate, because the reporting rule is *worst-served*, not *average* (PRD FR-6.2). Making the aggregate unavailable in the schema means it cannot be reported by accident.

4. Stage-2 storage

Database	PostgreSQL 15+
Extensions	pgcrypto (hashing), btree_gist (temporal exclusion)
Partitioning	ASSESSMENT and AUDIT_EVENT by month — one assessment per patient per minute is roughly 500k rows/month at 200 visits/day
Time series	Native timestampz; TimescaleDB optional and not required
Integration	Read-only HL7v2 ADT and FHIR R4 Observation ingest into OBSERVATION with source = monitor_feed. Read-only. The system never writes to the hospital record.
Sync	Device is the source of truth during a shift; assessments and audit records push on reconnect. Conflict policy: append-only, so there are no conflicts — the device's chain is grafted with its own segment identifier.

Required indexes:

```
CREATE INDEX ix_assessment_latest
  ON assessment (encounter_id, computed_at DESC);
CREATE INDEX ix_observation_series
  ON observation (encounter_id, observed_at DESC);
CREATE INDEX ix_audit_chain
  ON audit_event (shift_id, sequence);
CREATE INDEX ix_encounter_waiting
  ON encounter (shift_id) WHERE disposition = 'waiting';
CREATE INDEX ix_override_calibration
  ON override (direction, at DESC);
```

5. Authorisation

Enforced at Stage 2 with PostgreSQL row-level security. Stage 1 has a single device role and physical custody of the tablet is the access control — stated honestly rather than dressed up as a security model.

ROLE	ENCOUNTER	OBSERVATION	ASSESSMENT	OVERRIDE	AUDIT_EVENT	PROTOCOL	FAIRNESS
triage_nurse	C R U ¹	C R	R	C R	append only	R	—
charge_nurse	R U ¹	C R	R	C R	append only	R	R
physician	R U ²	C R	R	C R	append only	R	R
administrator	—	—	R ³	R ³	R	C R U	R
auditor	R ³	R ³	R ³	R ³	R	R	R
system	C R U	C R	C	—	append only	R	C

¹ own department, current shift only ² including disposition ³ de-identified projection only

Non-negotiable grants:

```
REVOKE UPDATE, DELETE ON audit_event FROM PUBLIC;
REVOKE UPDATE, DELETE ON assessment FROM PUBLIC;
GRANT INSERT, SELECT ON audit_event TO app_all_roles;
```

ASSESSMENT is insert-only for the same reason **AUDIT_EVENT** is: a mutable score history would make the central claim of the product — that priority is a trajectory, not a verdict — unverifiable.

Administrators cannot read identified encounter rows. Fairness oversight requires subgroup statistics, not individual patients, and the schema gives administrators exactly what oversight needs and nothing more.

Row-level policy for the nurse role:

```
CREATE POLICY nurse_current_shift ON encounter
  FOR ALL TO triage_nurse
  USING (shift_id = current_setting('app.shift_id')
        AND department_id = current_setting('app.department_id'));
```

6. Retention and de-identification

DATA	RETENTION	AT EXPIRY
ENCOUNTER (live)	24 h on device	Purged
OBSERVATION	24 h on device	Purged
ASSESSMENT	90 days identified, then de-identified	Exact age → age band; complaint text → class only; timestamps → 15-min buckets
AUDIT_EVENT	7 years	De-identified at shift close; the hash chain is recomputed over the de-identified payloads and both chain roots are retained so the transformation is itself auditable
OVERRIDE	7 years, de-identified	Retained for calibration
FAIRNESS_SNAPSHOT	Indefinite	Already aggregate, never identified
PROTOCOL	Indefinite	Superseded versions retained — an assessment must always be interpretable under the rules in force at the time

Regulatory frame. Primary: **India's DPDP Act 2023** — purpose limitation, data minimisation, storage limitation, and the fiduciary duties that follow from processing health data. The same design satisfies **HIPAA §164.312(b)** audit controls and §164.514 de-identification, and **GDPR Art. 9(2)(h)** processing for health care with Art. 5(1)(c) minimisation, without modification.

The reason one design satisfies three regimes is that Stage 1 collects no direct identifier at all. Compliance is cheap when there is nothing to protect, and that is an architectural choice made at the first schema decision rather than a legal exercise performed afterwards.

What a clinician override must legally record, and does: who (pseudonymous staff ID), when (timestamp), what the system recommended (band, confidence, index, interval), what the clinician decided (band), what the clinician was looking at (full inputs and derivation snapshot), and under which rule set (protocol version). This is the complete set required for an independent reviewer to reconstruct the decision, and it is exactly what §3.5 stores.

7. Stage-1 JSON shapes

The literal contents of `localStorage`, so an agent implementing `storage.js` has no ambiguity.

```
// key: "ptai.session.v1"
{
  "shiftId": "SH-20260823-A",
  "departmentId": "ED-REF-01",
  "startedAt": "2026-08-23T08:00:00+05:30",
  "protocolVersion": "v1",
  "engineVersion": "1.0.0",
  "mode": "NORMAL",           // NORMAL | SURGE | DEGRADED | SURGE_DEGRADED
  "nextEncounterSeq": 21,
  "nextEventSeq": 149,
  "staffId": "STAFF-04"
}

// key: "ptai.encounters.v1"
{ "PT-0007": { /* ENCOUNTER, §3.1 */ } }

// key: "ptai.observations.v1"
{ "PT-0007": [ /* OBSERVATION[], ascending by sequence, §3.2 */ ] }

// key: "ptai.assessments.v1"    - ring buffer, last 20 per encounter
{ "PT-0007": [ /* ASSESSMENT[], ascending by computed_at, §3.3 */ ] }

// IndexedDB: db "ptai", store "audit", keyPath "sequence", index "byEncounter"
{ /* AUDIT_EVENT, §3.7 */ }
```

Storage failure is not a failure state. If `localStorage` is unavailable — private browsing, full quota, a locked-down tablet — the application runs entirely in memory and displays a `NOT_PERSISTING` token in the header. Every read and write is wrapped in `try/catch` and the board renders correctly with no stored value. A triage board that refuses to start because it cannot write to disk is a triage board that is not there when the department needs it.

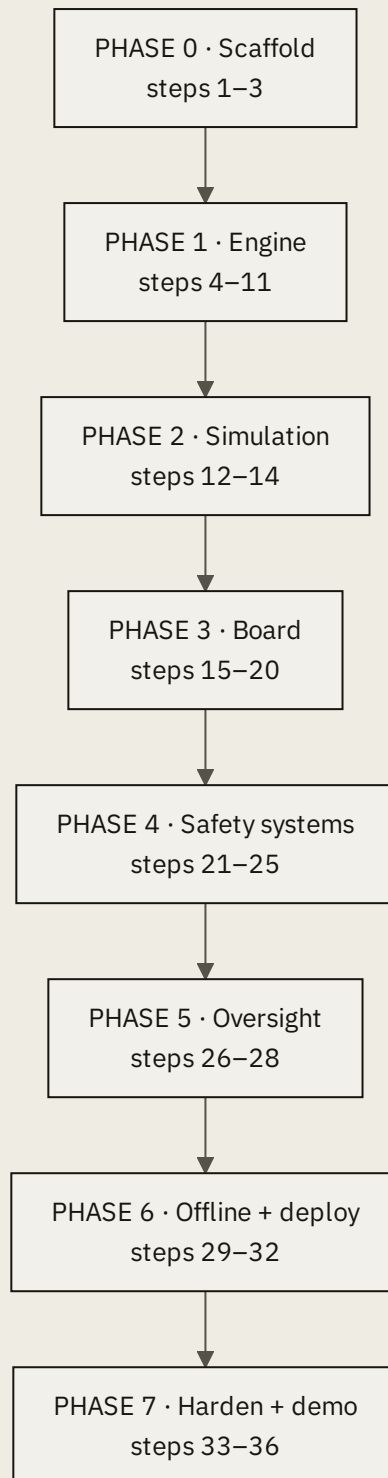
8. Engine ↔ storage contract

RULE	RATIONALE
<code>engine/</code> never touches storage, the DOM, or <code>Date.now()</code>	Determinism (PRD NFR-8) and testability. <code>now</code> is always an argument.
An ASSESSMENT is written on every tick for every waiting encounter	The trajectory is the product
Assessments are capped at the last 20 per encounter on device; older ones survive in the audit log	Bounded memory without losing the record
An AUDIT_EVENT of type SCORE is written only on band or confidence change , not on every tick	20 patients × 60 ticks/hour × 8 hours = 9,600 events/shift if unfiltered; change-only reduces this by roughly 95% while losing nothing a reviewer needs
Every write to AUDIT_EVENT recomputes and verifies the chain head first	A broken chain surfaces immediately, not at export
Export produces JSON and CSV, both readable without the application	PRD FR-4.4

Standards referenced: [ESI Handbook v5, 2023](#) · [RCP NEWS2](#) · [FDA CDS Software guidance, revised Jan 2026](#)

06 Implementation Plan Thirty-six ordered steps, each with an exit test	
Purpose	A chained, ordered development sequence for an AI coding agent or a human. Each step names its inputs, its deliverable, and the test that says it is finished. Do not start a step until the previous step's exit test passes.
Depends on	All of 01–05
Revision	1.0
0. How to use this document	
If you are an AI coding agent: read 01-PRD.md , 02-TRD.md , 04-UIUX-BRIEF.md in full before writing a line. Then work one step at a time, in order. At the end of each step, run the exit test and stop. Do not batch steps. Do not skip to the interesting part. The three rules that override everything else: 1. No build step, no framework, no runtime dependency. (TRD §1–2) 2. The prohibited-design list in 04-UIUX-BRIEF.md §2.1 is binding. No purple, no gradient, no shadow, no Inter, no icon library, no cards. 3. engine/ is pure. No DOM, no storage, no Date.now() . now is always a parameter. Ordering principle: the engine is built and tested before anything is rendered. This is the opposite of the usual instinct, and it is deliberate. The engine is the product; the board is a view of it. Building the UI first produces an interface that shapes the clinical logic to fit it, which is how safety-critical software goes wrong.	

1. Sequence overview



PHASE	STEPS	DELIVERABLE AT THE END
0 · Scaffold	1–3	Repo, tokens, a page that renders nothing but is correctly styled
1 · Engine	4–11	A tested, pure scoring engine with abstention. No UI at all.
2 · Simulation	12–14	20 encounters advancing on a clock, logged to console
3 · Board	15–20	The queue and inspector rendering live
4 · Safety	21–25	Override, audit chain, surge, degraded, alerts
5 · Oversight	26–28	Fairness monitor, export, reassessment prompts
6 · Offline	29–32	Service worker, manifest, Vercel, GitHub
7 · Harden	33–36	Accessibility, performance, design audit, demo rehearsal

PHASE 0 · Scaffold

Step 1 — Repository skeleton

Build: the directory tree in TRD §3, including `tests/` and `docs/`. Application files are created empty; `.gitignore` takes its contents verbatim from TRD §3.1 and `vercel.json` from TRD §10. `docs/07-DECISION-LOG.md` is created here with its index table and any decisions already taken. **Preserve:** every file already in the repository. TRD §3 is the required tree, not an exclusion list. **Do not:** run `npm init` at the repository root. The only package manifest is `tests/package.json`, and it is created in Step 11 when the first test needs Playwright — `node:test` needs no manifest at all. **Exit test:** every path in TRD §3 exists; no pre-existing file was deleted or moved; `.gitignore` matches §3.1 verbatim.

Step 2 — Design tokens

Build: `assets/css/board.css` containing **only** the token block — every custom property from UIUX §3.1, §3.2, §3.3, §4.1, §4.2, §5.1, §5.2, plus the `@media (prefers-color-scheme: dark)` override block and the Google Fonts `@import` for IBM Plex Sans, Sans Condensed, Mono and Serif. **Exit test:** open a scratch page that prints one swatch per token. Every token resolves. No literal hex appears below the token block. Compute every pair in the UIUX §3.4 table and confirm the measured ratio matches the value printed there — the table states measurements, so a mismatch means either the token or the table is wrong, and both must be reconciled before Step 3.

Step 3 — Page shell

Build: `index.html` with the semantic region skeleton from UIUX §5.3 (`<header>`, the grid, `<table>` with `<caption>` and `<thead>`, inspector `<aside>`, `<footer>` colophon). `assets/js/util/dom.js` (`el`, `on`, `text` — thirty lines, no library). `main.js` importing nothing yet. **Exit test:** the page loads with the correct grid proportions at 1280×800 and 1024×768, the colophon renders, and the diff contains zero `box-shadow`, zero `border-radius` above 2 px, and zero gradients.

PHASE 1 · The engine (no UI)

Everything in this phase is pure functions. Nothing renders. Resist the urge to look at it.

Step 4 – Protocol file

`assets/data/protocol.v1.json` ships complete in this repository. Do not author it, do not regenerate it, and do not "improve" its structure — it is clinical content under hospital governance, and its shape is documented in TRD §12.

Build: nothing. Read TRD §12 in full and verify the shipped file against it. **Exit test:** the file parses; all 14 rules carry `id`, `label`, `population`, `condition`, `action`, `alert`, `rationale`, `source`; every complaint class in `presentation.classes` has at least one entry in `resolvingQuestions`; every band array in `physiology` is contiguous with no gaps and no overlaps; every `table` path referenced by an `aboveAgeBandCeiling` operator resolves.

If verification fails, **stop and report** — a defect here is a clinical-content defect, not a coding task.

Step 5 – Age banding

Build: `engine/physiology.js` — `ageBand(ageValue, ageUnit)` returning the enum from Backend Schema §3.1.

Why first: every scoring path branches on the band *before* reaching any threshold table. Getting this wrong is the single highest-consequence bug available in this codebase. **Exit test:** unit tests at every boundary — 27 d / 28 d, 11 mo / 12 mo, 2 y / 3 y, 15 y / 16 y, 64 y / 65 y, 79 y / 80 y. Null age returns `adult` with the estimated flag.

Step 6 – Layer 0, hard-rule gate

Build: `engine/rules.js` — `evaluateRules(encounter, observation, protocol)` returning `RULE_FIRING[]`. A generic condition evaluator over `{ field, op, value }` supporting `<`, `>`, `<=`, `>=`, `in`, `=`, and the paediatric formula operator `lt_pediatic_hypotension`. **Exit test:** every rule in TRD §4.1 fires on a crafted positive case and stays silent on a crafted negative. `PIN_P1` always sets `modelLockedOut = true`. **Additionally assert that no `population: adult` rule ever fires for a paediatric encounter** — construct a 3-year-old at SpO₂ 87 and RR 7 and confirm `RULE-RESP-01` and `RULE-RESP-02` stay silent while `RULE-PAED-04` and `RULE-PAED-05` both fire.

Step 7 – Layer 1, physiology

Build: the adult NEWS2 table (TRD §4.2.1) **exactly as published**, the paediatric age-band tables (§4.2.2), and the obstetric shift (§4.2.3). Return `{ score, perParameter[], missing[], singleParameterThree }`. **Exit test:** transcribe the published NEWS2 chart into a fixture and assert every cell. Assert the single-parameter-3 flag independently of the aggregate. Assert paediatric hypotension at ages 1, 5 and 10 against `70 + 2 × age`. **Assert that each of the three population paths scores all seven parameters and can reach exactly 20**, against `physiology.parameterMaxima.populationPaths` — a path that cannot reach 20 is missing a parameter. Assert paediatric SpO₂ boundaries at 91/92, 94/95 and 96/97, and the bradypnoea floor for every age band. **Watch for:** the boundary values. NEWS2 scores RR 20 as 0 and RR 21 as 1. Off-by-one here is a clinical error, not a cosmetic one.

Step 8 – Layer 2, presentation

Build: `engine/presentation.js` — the complaint-class table and modifier set from TRD §4.3. Every modifier that fires returns its label and points. **Exit test:** PT-0007's profile (61 M, `abdominal_pain`, `radiating` qualifier, **not** diaphoretic) returns base 6 + 7 with PM-AB-01 labelled — the modifier fires on either limb, not both. `unknown` class returns base 8. `chest_pain` with every modifier firing returns 20, **and the derivation still lists all four modifiers with a clamped flag**. Every qualifier chip in Backend Schema §3.1 is referenced by at least one modifier — an unused chip is a capture field that cannot affect a score.

Step 9 – Layer 3, hazard and drift

Build: `engine/hazard.js` — time hazard by band and drift from the observation series (TRD §4.4). A single observation returns drift 0 **and** sets `singleReading = true`. **Exit test:** reproduce the worked example in TRD §4.4 to the stated value. A single-observation encounter never produces non-zero drift.

Step 10 — Uncertainty and abstention

Build: `engine/uncertainty.js` — evidence completeness, interval half-width including `driftUncertainty`, the **five** confidence levels, `expectedInformationGain`, and deterministic question ranking (TRD §4.6). **Exit test:** removing each input in turn reduces completeness by its documented weight. An interval straddling a boundary at a 60:40 split yields `UNRESOLVED` with two candidate bands and exactly one question. `E < 0.45` yields `INSUFFICIENT`. **A question whose `alreadyAnsweredWhen` holds is never offered** — assert with PT-0007 that RQ-ABDO-01 is suppressed and RQ-ABDO-02 is returned instead. When every question for the class is suppressed, the state is `UNRESOLVABLE` with reason `all_questions_already_answered` — **never `UNRESOLVED` with a null question**; assert that combination throws. Assert the same for a question scoring below `minimumInformationGain`, with reason `no_question_above_information_threshold`. Assert question ranking is stable across 100 runs with shuffled input key order.

Assert every fixture in `protocol.v1.json` → `workedExamples` — PT-0007-abstention (suppression, chosen question, EIG 0.7747), RQ-ABDO-01-if-not-suppressed (0.8663), all five `driftUncertainty` cases including the zero at one observation and the cap at two, and both `consistencyCases` — **identical drift magnitude, four readings, differing only in agreement, must produce 1.63 against 4.67**. `monotonicFraction` is 0 below three observations. These fixtures exist so that a numeric value drifting out of step with a formula fails a test rather than blocking a later step; treat a failure here as a defect in whichever of the two is wrong, and report rather than adjusting the fixture to match the code.

Step 11 — Banding, tie-break, and the contract assertion

Build: `engine/bands.js` and `engine/index.js` — combine the layers (TRD §4.5), apply the asymmetric tie-break, apply hard-rule and **presentation floors** in the order at §12.4 steps 4–7, set `bandSetBy`, assemble the `Assessment` object (TRD §4.7), and **throw** on any of the seven contract violations listed there. **Note on ordering.** This step's golden snapshots need the cohort, and the cohort ships complete in the repository (Step 12 builds only its loader). Read `assets/data/cohort.json` directly here; a minimal inline reader is acceptable, and Step 12 replaces it with the real interpolating loader.

Exit test — this is the gate for the whole phase: - Every `expect` block in `cohort.json` passes — band, `bandSetBy`, confidence, and rules fired, at each stated minute. These are clinical intent and are authoritative. If one fails, report it; do not edit the cohort to match the engine. Note that `bandIn` / `confidenceIn` accept any member of the listed set: they are used where the exact value is not the clinical point, and asserting a single value there would make the fixture brittle rather than informative. - **Abstention fires only at P1/P2 and P2/P3.** Assert that an interval straddling P3/P4 or P4/P5 tie-breaks upward and reports `PROBABLE` with no resolving question. PT-0012 and PT-0018 are the fixtures. - Golden-file snapshot of all 20 cohort encounters at t=0, t=30 and t=60. Golden files are **generated** from the engine and then reviewed once against the `expect` blocks — they capture detail the expectations do not, and they exist to detect future drift, not to define correctness. - **Property test, 10,000 random encounters:** every output has a `confidence`; every `PIN_P1` yields `P1`; `interval_low ≤ priorityIndex ≤ interval_high`; identical inputs produce byte-identical output. - **Assert the floor arithmetic directly:** an encounter with `L1 = 1` and `L2 = 20` and `L3 = 20` reaches only PI 56.5 by the continuous score, and lands at P2 *only* via the presentation floor with `bandSetBy = "presentation_floor"`. This is the property that makes atypical presentations catchable at all. - Manually verify the named cases against clinical intuition: PT-0004 and PT-0020 reach P2 on presentation with unremarkable vitals, PT-0007 abstains at P2/P3, PT-0011 fires RULE-PAED-03 on the preschool ceiling, PT-0013 pins to P1 with no identity, PT-0019 returns `INSUFFICIENT` and still queues at P2.

Do not proceed to Phase 2 until every test in Step 11 is green. Every downstream phase assumes the engine is correct, and an engine bug found in Phase 5 costs ten times what it costs here.

PHASE 2 · Simulation

Step 12 — Cohort loader

assets/data/cohort.json ships complete in this repository, alongside **protocol.v1.json**. It is clinical content: 20 synthetic encounters, each with keyframed trajectories and an **expect** block stating the band, mechanism and confidence it must produce. Do not author it and do not adjust its values to make a test pass.

Build: **sim/cohort.js** only — load the file, linearly interpolate between keyframes, hold the last keyframe thereafter, and project each encounter into the flat field view the engine reads. **Exit test:** interpolation at a keyframe returns the keyframe exactly; between keyframes returns the linear value at field precision; after the last returns the last. **unobtainable** survives projection as **unobtainable**, distinct from null. PT-0002 at t=45 reproduces its final keyframe.

Step 13 — Clock

Build: **clock.js** — a simulation clock with run/pause and 1× / 10× / 60× speeds, emitting a tick. **Date.now()** appears nowhere else in the codebase. **Exit test:** at 60×, one wall-clock minute advances one simulated hour. Pause freezes every derived value.

Step 14 — Tick loop, headless

Build: wire clock → advance trajectories → score every encounter → sort by index. Log the ordered board to the console each tick. **Exit test:** run 90 simulated minutes headless. PT-0002 rises past PT-0010. No encounter's **lastRecomputeAge** exceeds 60 s. Recompute of 20 encounters completes in under 100 ms.

PHASE 3 · The board

Step 15 — Queue table

Build: **render/board.js** — the fixed-column table from UIUX §5.4, one row per encounter, band chip, vitals with units, wait, confidence indicator. Keyed by encounter ID, rows reused across renders. **Exit test:** 20 rows render; vitals align vertically down the whole list; **---** shows for unobtainable; no zebra striping; every numeric cell uses **tabular-nums**.

Step 16 — Row states

Build: the eight states in App Flow §4.2 — normal, selected, moved, overdue, abstaining, rule-pinned, overridden, collapsed. **Exit test:** each state is reachable and each carries hue **and** text token **and** glyph. Screenshot every state.

Step 17 — Movement and the two animations

Build: rank-change detection, the movement line with its cause string, and the two animations from UIUX §7 — value flash and row translate. Nothing else animates. **Exit test:** at 60× the movement is followable by eye. **prefers-reduced-motion: reduce** substitutes the persistent marker. **grep -c "animation\|transition"** in the CSS returns only the two.

Step 18 — Inspector

Build: **render/inspector.js** — the full layout in App Flow §5: index with confidence band, confidence statement, resolving question, the complete layered derivation, vital trends, reassessment time, engine and protocol footer. **Exit test:** every number on the panel is traceable to a derivation line. Missing inputs appear as stated line items, never as absences. The empty state shows the board summary.

Step 19 — Charts

Build: `render/charts.js` — the confidence band, the sparkline, and the fairness bars as hand-authored inline SVG (UIUX §6.3, §6.5, §6.6). **No charting library.** **Exit test:** the confidence band's marker sits at the point estimate and the span matches the interval. Sparklines render with six to eight points, 1 px stroke, no axes.

Step 20 — Arrival and reassessment sheets

Build: `S3` and `S4` per App Flow §6 and §7. Numeric keypad, not the OS keyboard. Per-field `UNOBTAINABLE` toggles. ACVPU the only required field. **Exit test:** time a full capture — it must complete in under 90 seconds on a touch device. A patient with no age, no sex, no complaint and no obtainable vitals still produces a scoreable record. No scrolling required at 1280×800.

PHASE 4 · Safety systems

Step 21 — Override

Build: drag-to-reposition and `Shift+↑ / ↓ + Enter`. **No dialog. No justification field.** The override commits before anything else happens. **Exit test:** override completes in one interaction. The inspector shows the engine band struck through beside the nurse band — never hidden.

Step 22 — Audit log and hash chain

Build: `audit.js` — IndexedDB append-only store, the record shape in Backend Schema §3.7, SHA-256 chaining via `crypto.subtle`, chain verification on every write. `SCORE` events written on band-or-confidence change only, not every tick. **Exit test:** an override writes a record containing the complete state from App Flow §8. Verification passes on a clean chain and fails on a hand-tampered record. An 8-hour simulated shift produces under 500 events, not 9,600.

Step 23 — Audit drawer and export

Build: `S5` per App Flow §8, with JSON and CSV export. **Exit test:** the exported file is readable and reconstructible without the application. Chain verification passes on the export.

Step 24 — Surge mode

Build: `sim/surge.js` — 3× arrival injection, automatic mode entry and exit, top-five retention, row collapse, the 33% reassessment compression, the stated trigger banner, and a `MODE_CHANGE` audit record carrying the measured values. **Exit test:** injecting 18 arrivals/hour enters surge within 15 simulated minutes. Ranks 6+ collapse. The banner states the actual rate and multiplier, not a generic message. Exit requires 10 continuous minutes below 2×.

Step 25 — Degraded mode and the emergency alert

Build: the instrument-unavailable path — all instrument vitals to unobtainable, widened intervals, and a banner that **names which bands can no longer be discriminated and for how many patients**. Then `S8`, the absolute-emergency alert bar, firing on any `PIN_P1`, with logged acknowledgement that does not clear the pin. **Exit test:** in degraded mode more patients enter abstention, the queue keeps ordering, and the banner counts them correctly. PT-0013 fires the alert on load; acknowledging logs the event and leaves the pin. No path exists to sort a pinned row below another row.

PHASE 5 · Oversight

Step 26 — Fairness monitor

Build: `fairness.js` and `S6` per App Flow §9 — subgroup distributions by sex, age band and language; upgrade-after-triage as the undertriage proxy; the worst-served subgroup named **in a sentence** at the top; drift against tolerance. **Exit test:** the headline is a specific sentence naming a subgroup and a multiple. **No aggregate fairness number appears anywhere on the panel.** Flagged subgroups are reachable to their encounters.

Step 27 — Reassessment prompts

Build: the overdue state driven by `reassessMinutes` per band, escalating visibility over time. **Exit test:** a P3 waiting 31 minutes enters overdue. An un-reassessed patient becomes more visible, never less, and is never dropped from the board.

Step 28 — Simulation console

Build: `S9` per App Flow §12, visibly marked `PROTOTYPE CONTROLS — NOT PART OF THE CLINICAL PRODUCT`. **Exit test:** every control in the App Flow §12 table works. The console is unmistakably scaffolding and could be deleted in one commit.

PHASE 6 · Offline and deployment

Step 29 — Service worker

Build: `sw.js` — precache manifest covering HTML, CSS, all JS modules, both JSON files, and the four Plex font files; cache-first; versioned cache name with cleanup on activate. **Exit test:** **serve the app over HTTP, load it once, disable the network, hard-reload. It works completely.** This is the Stage-1 product claim (TRD §1); it is not optional and it is not a nice-to-have. Note that `file://` is not the test — service workers do not register on that scheme.

Step 30 — Manifest and tablet install

Build: `manifest.webmanifest` — landscape orientation, standalone display, the ground colour as theme colour, maskable icon. **Exit test:** installs to an Android tablet home screen and opens without browser chrome in landscape.

Step 31 — GitHub

Build: `README.md` — what it is, what it is not, the 10-step demo path from App Flow §13, how to run locally, links to all six docs, and an explicit "synthetic data only, not for clinical use" notice. Then initialise and push.

```
git init && git add -A
git commit -m "PatientTriage.ai: engine, board, safety systems, offline shell"
git branch -M main
git remote add origin git@github.com:<org>/patienttriage-ai.git
git push -u origin main
```

Exit test: a clean clone plus `python3 -m http.server 8000` runs the application with no install step.

Step 32 — Vercel

Build: import the repo. Framework preset `other`, build command empty, output directory `.`. Confirm the `vercel.json` headers from TRD §10 apply. **Exit test:** the production URL loads in under 2 s, the service worker registers over HTTPS,

security headers are present, and a pull request produces a working preview URL.

PHASE 7 · Hardening and the demo

Step 33 — Accessibility pass

Build: to UIUX §9 — keyboard path, focus rings, ARIA live regions, table semantics, 200% zoom. **Exit test:** complete the full demo path with the mouse unplugged. axe reports zero violations. Simulate deuteranopia and confirm every state is still separable.

Step 34 — Performance pass

Exit test: every budget in TRD §7 met, measured — not estimated. Total transfer ≤ 120 kB gzipped. Board recompute ≤ 100 ms at 60 encounters. Zero long tasks over 50 ms during a tick.

Step 35 — Design audit

Exit test: work the UIUX §11 checklist line by line against the full diff. Then the harder test: **show the deployed URL to three people and ask whether it looks AI-generated.** If any of them hesitates, the answer is in §2.1 — find which line was crossed.

Step 36 — Demo rehearsal

Exit test: run the 10-step path in App Flow §13 end to end, timed, three times. It must fit five minutes with room to be interrupted. Step 10 — network off, reload, still working — is the closing move and must be rehearsed until it is boring.

2. Definition of done

The prototype is finished when all of the following are simultaneously true:

- [] 20 synthetic encounters score correctly against the golden file
- [] An ambiguous case (PT-0007) abstains and names exactly one resolving question
- [] A paediatric case (PT-0011) scores on age-band tables, not adult tables
- [] A geriatric atypical case (PT-0004) reaches P2 from presentation risk with unremarkable vitals
- [] A zero-history case (PT-0013) is admitted and pinned with no identity whatsoever
- [] 3× surge switches the board and states its trigger with real numbers
- [] **No score anywhere renders without a confidence indicator**
- [] A clinician override commits in one interaction and produces a complete, chain-verified audit record
- [] The fairness monitor names a worst-served subgroup in a sentence
- [] **The application works with the network off**
- [] Every budget in TRD §7 is met by measurement
- [] The UIUX §11 checklist passes on the full diff
- [] Live on Vercel from `main`, and a clean clone runs with no install

3. Sequencing rationale

Four ordering decisions are load-bearing, and an agent tempted to reorder should read this first.

The engine precedes the interface. Phase 1 produces no visible output at all. This feels wrong and is correct: the engine is the product and the board is a view of it. Building the view first produces an interface that quietly reshapes the clinical logic to suit its layout, which is a well-worn path to unsafe software.

Age banding is step 5, before any scoring. It is the highest-consequence single function in the codebase. A fever of 38.5 °C means something different in a 3-year-old than in a 75-year-old, and a system that reaches an adult threshold table before it has established the age band has already made the error.

Uncertainty is built with the engine, not bolted on. Abstention is a first-class output of `engine/index.js` (Step 10, before banding in Step 11), not a UI treatment applied to a low score. A system that computes a confident answer and then decorates it with a caveat is not the same system as one that can genuinely decline — and only the second one is honest.

Offline is Phase 6, not "later". It is scheduled before hardening because it is a product requirement rather than an optimisation. If it slips past Phase 6 it will not happen, and Stage 1 — one tablet, no internet, a district hospital at 2 a.m. — is the entire deployment story.

4. Estimated effort

For a three-person team, assuming the docs are read first.

PHASE	EFFORT	NOTES
0 · Scaffold	0.5 day	
1 · Engine	2.5 days	The longest phase, and the one to protect
2 · Simulation	1 day	Cohort authoring dominates
3 · Board	2 days	
4 · Safety	1.5 days	
5 · Oversight	1 day	
6 · Offline + deploy	0.5 day	
7 · Harden + demo	1 day	
Total	≈ 10 working days	

Parallelisation: after Step 11, Phase 3 (board) and Phase 4 (safety systems) can run on separate branches, because both consume the engine and neither modifies it. Phases 1 and 2 must be sequential.

5. What is deliberately not built

Naming these in advance stops them from being rediscovered as scope creep at day seven.

NOT BUILT	WHY NOT, AND WHEN IT ARRIVES
Authentication and role management	Stage 1 is one tablet under physical custody. Roles arrive with the Stage-2 server, where Backend Schema §5 already specifies them.
HL7 / FHIR integration	Stage 2 ("The Background Listener"). Stage 1's value must be provable without it, and if it were built now it would become a dependency the pilot cannot shed.
A trained ML classifier	The engine is a calibrated rule-and-weight system, deliberately. There is no labelled ED dataset with the right outcome — post-triage deterioration — and a classifier trained on the wrong label would be less safe and far less explainable than transparent weights. Weights are a hypothesis to be calibrated against pilot data, and they are honest about being one.
Multi-hospital capacity network	Stage 3 ("The Traffic Director").
Native mobile applications	The web app installs to a tablet home screen and works offline. A native shell adds a build pipeline and an app store for no clinical gain.
Any analytics or error-reporting service	A clinical tool that phones home is a clinical tool that leaks. TRD §11.

08

Brief Coverage

Where each item in the problem statement is answered

Where each item in the Round 2 problem statement is answered. Written so a reader can check the submission against the brief without reading all of it.

Minimum prototype expectations

EXPECTATION	WHERE	STATUS
Triage scoring on 15–20 simulated records	assets/data/cohort.json — 20 encounters	Met
At least one ambiguous presentation	PT-0007 · 61 M, "stomach ache... goes up into my shoulder". Abstains across P2/P3, names one resolving question	Met
At least one paediatric case	PT-0011 · 3 y, HR 168 against the preschool ESI ceiling of 120. PT-0015 · 6 mo, where 184 bpm is 4 above the infant ceiling and unremarkable in a toddler	Met
At least one geriatric case	PT-0004 · 82 F, afebrile sepsis reaching P2 on collateral history with unremarkable vitals	Met
At least one zero-history patient	PT-0013 · unidentified, unresponsive, no name, no age, no complaint — scored and pinned to P1	Met
Behaviour under 3× surge	sim/surge.js; entered on measured trailing-15-minute arrival rate, not a button. Top five retained, rest collapse, reassessment intervals compress 33%	Met
Uncertainty explicit — no score without a confidence indicator	Five confidence states; engine/index.js throws if an assessment would return without one; database CHECK mirrors it. Property-tested over 10,000 random encounters	Met
At least one clinician override, and what is logged	Drag or Shift+↑+Enter, one gesture, no dialog. Writes an append-only SHA-256-chained record with the engine's band, confidence, index, interval, every input and the full derivation snapshot	Met

Real-world complexities

COMPLEXITY	ANSWER
Overlapping / ambiguous symptoms; under-reported pain	Layer 2 presentation risk with age and sex modifiers. PM-AB-01 catches the "stomach ache" that is an MI; PM-CP-02 exists because women's cardiac presentations are under-recognised
Age-banded thresholds — 38.5 °C means different things at 3 and 75	Age band is resolved before any threshold table is selected. Separate adult NEWS2, paediatric age-band, and obstetric paths, each scoring all seven parameters to a maximum of 20 — asserted in test, because a path that cannot reach 20 is missing a parameter
Data quality varies; half have prior records	PRD §7.1. Zero-history is the default path, not the exception; history arrives at Stage 2 as observations and flags, never as a separate scoring path
Explainable in seconds, by an interrupted clinician	Row meta line for the glance; inspector for the full derivation, every layer and contribution, on the same screen
Asymmetric under/over-triage cost	Explicit 8:1 cost matrix. Ties break upward and the interface says so. Presentation floors implement ESI decision point B so a dangerous presentation reaches P2 on normal vitals — without them the continuous score cannot exceed P3, which is proven arithmetically in TRD §4.3
Hospitals differ in scale and specialty	PRD §13. Every contestable clinical value is in a hospital-owned protocol file; the surge trigger is a multiple of that department's own baseline
Accountability, override, audit, regulation	Recommend-not-decide boundary; append-only hash-chained log; DPDP Act 2023 primary, satisfying HIPAA and GDPR by construction
Integration is rarely simple	Three stages, each delivering value alone. Stage 1 needs one tablet and no integration

Solutioning areas

AREA	WHERE
Data strategy	TRD §4.2–4.4 · uncertainty weights TRD §4.6 · Backend Schema §3
Decision model and its own uncertainty	TRD §4 — four layers, hybrid rules-and-weights, five confidence states, abstention as a first-class output
Workflow design	App Flow §4–§10 — surfacing, override capture, surge versus quiet shift
Safety-first defaults and deterioration monitoring	PRD §4 (P1–P8) · TRD §4.1, §4.4 · reassessment intervals per band, and drift-driven re-ranking
Adoption and change management	PRD §12
Patient data protection	Backend Schema §6 — minimisation as architecture; Stage 1 collects no direct identifier
Scalability	PRD §13

Stated assumptions

Regulatory jurisdiction **India**, **DPDP Act 2023**, designed to satisfy HIPAA §164 and GDPR Art. 9 without change. Five-level scale, ESI v5 aligned. Baseline 6 arrivals/hour, surge at 3×. Undertriage:overtriage 8:1. Retention 24 h for encounters, 7 years de-identified for audit. All in PRD §7, all in the protocol file.

Deliberately not built

The **Overflow Valve** (FR-7.1, FR-7.2) — partner-clinic slots and city-network ambulance redirection. It depends on a live capacity feed that does not exist to integrate against and cannot be honestly simulated. PRD §9 states the reasoning. Also out of scope: EHR/HL7 integration, authentication and roles, trained ML weights, regulatory submission artefacts.

A

AGENTS.md

Context file for AI coding agents

Works as `AGENTS.md`, `CLAUDE.md`, `.cursorrules`, or a Lovable knowledge entry. Copy it in verbatim.

Project

PatientTriage.ai — a decision-support layer over an emergency-department waiting room. It scores triage priority from sparse arrival data, recomputes continuously, abstains when it cannot discriminate, and hands every final call to the nurse.

Full specification lives in `docs/01-PRD.md` through `docs/06-IMPLEMENTATION-PLAN.md`. **This file is a summary, not a substitute.** Read `02-TRD.md` and `04-UIUX-BRIEF.md` in full before writing code.

Hard constraints — violating any of these is a rejected change

Architecture

1. **No build step.** No Vite, Webpack, Rollup, Parcel, esbuild, or npm script in the deploy path. The files in the repo are the files that are served — nothing is compiled or generated. Offline operation comes from the service worker after one HTTPS load, or from a packaged WebView; `file://` is not the supported path (TRD §1).
2. **No framework.** No React, Next.js, Vue, Svelte, Angular, Alpine, htmx, jQuery.
3. **No runtime dependencies.** Zero. Native ES modules only.
4. **No Tailwind, no shadcn/ui, no Bootstrap, no CSS-in-JS.** One hand-written stylesheet with a custom-property token block at the top.
5. **No charting library.** No Chart.js, D3, Plotly, Recharts. Charts are hand-authored inline SVG.
6. **No icon library and no emoji.** No Lucide, Feather, Heroicons, Font Awesome, Material Icons. Five typographic marks plus two hand-drawn SVGs.
7. **No date library, no HTTP client, no state library, no polyfills.** `Intl`, `Date`, `fetch`.

If a task appears to need one of these, the task has been misread. Re-read `docs/02-TRD.md`.

Engine purity

`assets/js/engine/` contains pure functions only.

- No DOM access.
- No storage access.
- **No `Date.now()`.** `now` is always a parameter.
- Same inputs → byte-identical output, always.

This is what makes the scores deterministic, reproducible, and testable, and it is a stated non-functional requirement, not a style preference.

Clinical safety invariants

These are enforced in code, not by convention. `engine/index.js` throws if any is violated.

INVARIANT

- | | |
|---|--|
| 1 | No assessment is ever returned without a confidence value. |
| 2 | A fired <code>PIN_P1</code> hard rule always yields band <code>P1</code> and <code>modelLockedOut = true</code> , and the engine may not emit any recommendation containing a wait. |
| 3 | <code>interval_low ≤ priorityIndex ≤ interval_high</code> , always. |
| 4 | Ties break upward. Undertriage costs 8× overtriage. |
| 5 | Age band is computed before any threshold table is selected. Never look up an adult table and correct for age afterwards. |
| 6 | Abstention is a real output state, with <code>band = null</code> , at least two <code>candidateBands</code> , and a <code>provisionalBand</code> . <code>UNRESOLVED</code> always carries exactly one resolving question — no exceptions, no null. Where no question can resolve the ambiguity the state is <code>UNRESOLVABLE</code> , which carries a <code>noQuestionReason</code> instead and instructs escalation. Never emit <code>UNRESOLVED</code> with a null question. |
| 7 | Override never requires a dialog or a justification field. <code>reason_chip</code> is nullable by design. |
| 8 | The audit log is append-only and hash-chained. No <code>UPDATE</code> , no <code>DELETE</code> . |

Design contract — binding

Full detail in `docs/04-UIUX-BRIEF.md` §2. The short version:

Never

Inter · Roboto · Poppins · Montserrat · Space Grotesk · purple/indigo/violet in any form · any gradient · any `box-shadow` · `border-radius` above 2 px · glassmorphism or backdrop-blur · cards · three-across feature grids · icon libraries · emoji · `fade-in-up` or any entrance animation · uniform `gap-4` spacing · stock imagery · marketing copy register.

Always

- **Type:** IBM Plex Mono (all numbers), IBM Plex Sans (prose), IBM Plex Sans Condensed (uppercase micro-labels), IBM Plex Serif (inspector derivation headings only).
- **Ground:** `#F2F0EA` warm paper. Structure carried by 1 px hairlines. Zero shadows anywhere.
- **Colour is a signal, never a surface.** Five hues, each bound to one meaning. P3 — the middle band — is **achromatic on purpose**.
- **Every state carries hue + text token + glyph.** Colour is never the sole carrier of meaning.
- `font-variant-numeric: tabular-nums` on every numeric element.
- **Two animations only:** a 140 ms value flash and a 180 ms row translate. Nothing else moves.
- **Unobtainable values render as** `--`, never `0`, `N/A`, `null`, or an empty cell.

Copy register

Terse, clinical, declarative. Every string states a fact and, where relevant, its consequence.

Write: *"Not obtained. Interval widened."* · *"Cannot discriminate P2 from P3."* · *"Tie broken upward. Undertriage cost 8:1."*

Never: *"No data available"* · *"Low confidence"* · *"Erring on the side of caution"* · anything with an exclamation mark.

File map

```
index.html · sw.js · vercel.json · manifest.webmanifest · .gitignore
docs/                                incl. 07-DECISION-LOG.md – append as you build
tests/  unit/ property/ golden/ e2e/ fixtures/ package.json  dev only, never served
assets/css/board.css                 tokens at top, no literals below
assets/data/protocol.v1.json          hospital-owned rules – NOT code
assets/data/cohort.json               20 synthetic encounters
assets/js/
  main.js state.js clock.js audit.js fairness.js
  engine/  index rules physiology presentation hazard uncertainty bands
  sim/     cohort surge
  render/  board inspector modes charts
  util/    dom fmt storage
```

One module, one responsibility. `main.js` wires; it contains no logic.

Working method

1. Follow `docs/06-IMPLEMENTATION-PLAN.md` **one step at a time, in order**. Do not batch. Do not skip ahead to the visible parts.
2. Run each step's exit test before starting the next.
3. The engine (Phase 1, steps 4–11) is built and fully tested **before anything renders**. This feels wrong and is correct.
4. When a clinical threshold is needed, it comes from `protocol.v1.json` — never a literal in code.
5. When unsure about a value, look it up in the docs. If it genuinely is not specified, `docs/DECISION-PROTOCOL.md` **governs**: decide on the conservative branch, log it, and proceed. Stop only for a clinical constant with no derivable basis, a missing safety gate, an absent content artefact, or an architectural property that makes a stated claim unreachable.

Domain vocabulary

TERM	MEANING HERE
Band	Priority level P1–P5, aligned to ESI 1–5
Priority Index	Continuous 0–100 score behind the band
Abstention	The engine declines to assign a band and names two candidates. <code>UNRESOLVED</code> asks one question; <code>UNRESOLVABLE</code> says questioning cannot help and escalates; <code>INSUFFICIENT</code> asks for missing observations.
Information gain	0–1 score for how much a question would resolve the current ambiguity. Below 0.25 the question is not worth the nurse's twenty seconds.
Provisional band	The safe upper band used for queue position while abstaining
Hard rule / LO	Deterministic hospital-owned criterion that overrides the model entirely
Drift	Rate of change of vitals across the observation series
Time hazard	Risk accrued from waiting, weighted by current band
Evidence completeness (E)	0–1 measure of how much input was actually obtained
Unobtainable	Attempted and failed — distinct from "not recorded"
Override	The nurse repositioning a patient. Always permitted, always logged, never blocked.
Worst-served subgroup	The fairness reporting unit. Aggregates are never reported.

Testing

- **Unit:** `node:test` . NEWS2 tables cell-by-cell against the published chart. Paediatric boundaries. Hypotension formula at ages 1, 5, 10.
- **Property:** 10,000 random encounters — every output has a confidence; `PIN_P1` always yields P1; the interval always contains the point estimate.
- **Regression:** golden-file snapshot of all 20 cohort assessments at t=0, t=30, t=60 min.
- **Integration:** Playwright — board renders, drift moves PT-0002, surge switches mode, override writes a chain-verified audit record.

No push to `main` without unit and property tests green.

Deployment

Static, from GitHub `main` , to Vercel. Framework preset `other` , build command **empty**, output directory `.` . There is nothing to configure because there is nothing to build.

Local: `python3 -m http.server 8000` — needed only because ES modules and service workers are blocked on `file://` .

B Decision Protocol

When a build agent decides, and when it escalates

Applies to	Any agent building from docs/06-IMPLEMENTATION-PLAN.md
Supersedes	The instruction in PROMPT-FOR-CODEX.md to "stop and ask" on any ambiguity
Revision	1.0

1. The default has changed

Earlier instructions told you to stop whenever the documents were ambiguous. That was right while the specification was thin. It is now the wrong default: the docs are dense, the remaining gaps are mostly mechanical, and each stop costs a round trip that is more expensive than the decision it protects.

New default: decide, log, and proceed. Stopping is the exception, and it must be earned.

Calibration from the blockers raised so far: of eight, **three** genuinely needed the clinical owner. The other five — repository structure, `.gitignore` contents, test paths, stale numeric literals, a doc-to-doc conflict with an obvious safe branch — you could have resolved yourself and logged.

2. The test

Ask one question:

Could getting this wrong change which band a patient is assigned — and is there no safe default?

Both must be true to stop. If the answer to either half is no, decide it yourself.

That second clause carries most of the weight. Most clinical ambiguities *do* have a safe default, and §3 tells you what it is.

3. The safe-default rule

The product already has a principle for deciding under uncertainty. Apply it to specification uncertainty too.

When two readings are defensible, take the one that is more conservative for the patient.

Concretely, prefer the option that:

- assigns the **higher** acuity, never the lower
- makes the interval **wider**, never narrower
- fires a rule **more** often, never less
- makes a patient **more** visible on the board, never less

- writes **more** to the audit log, never less
- **abstains** rather than asserts

This mirrors the 8:1 undertriage cost that governs the engine. An overtriaged specification decision is a nuisance; an undertriaged one is the failure mode the product exists to prevent. Where the safe branch is obvious under this rule, **you do not have a blocker — you have a decision.**

Worked: PM-AB-01 read **AND** in one document and **OR** in another. **OR** fires more often, so **OR** is the safe branch. That was a Tier 2 decision, not a stop.

4. Three tiers

Tier 1 — Decide and proceed. Log normally.

No clinical consequence. Do not mention these in your step report beyond the log entry.

- File and directory structure, naming, module boundaries
- **.gitignore**, config files, test layout, fixture organisation
- JSON shape for anything that is not a clinical value
- Stale literals that contradict a formula — **the formula wins**, fix the literal
- Doc-to-doc conflicts where one source is clearly more specific or more recent
- Anything the plan assigns to a later step that you need now — pull it forward
- Obvious typos, broken cross-references, missing plumbing

Tier 2 — Decide on the safe default, proceed, mark **PROVISIONAL**.

Clinical consequence exists, but §3 gives a defensible branch.

Log the entry with ****Status:** provisional**, add a row to the **Provisional Decisions** table at the top of **07-DECISION-LOG.md**, and keep going. These are reviewed in one batch at Step 33 — nothing is lost, and nothing blocks.

- A threshold specified in one place and contradicted in another
- A formula referenced but not defined, where a standard or a stated principle implies one
- A missing modifier, question, or field the surrounding structure clearly expects
- An enum that needs a new member to represent a real state
- A calibration that produces a documented case landing in the wrong band

For that last one: if a value must be invented, **derive it rather than guess it**. State the derivation in the log ("30% below the lower bound of the age-band normal"), so it can be checked rather than merely accepted.

Tier 3 — Stop and ask.

Only these. All four conditions of the tier apply: clinical consequence, no safe default, cannot be derived, and the answer is a judgement a clinician owns.

- **A clinical constant with no derivable basis** — a new red-flag threshold, a scoring band, a cost ratio
- **A safety gate that appears to be missing entirely**, where adding one is itself a clinical judgement (the paediatric SpO₂ gate was correctly this)
- **A whole content artefact that does not exist** and cannot be derived from what does (the protocol file, the cohort)
- **An architectural property that makes a stated product claim unreachable** (the presentation-floor arithmetic)

- **Anything where the conservative branch is also plausibly unsafe** — for example, a rule that would fire so often it would cause alert fatigue and degrade the response to real alarms

If you are unsure which tier applies: it is Tier 2. Tier 3 is a short list, and it is short on purpose.

5. Before you stop: the overthinking check

Run this before raising any Tier 3. If your tooling supports a subagent, spawn one with the prompt in §6 and give it a fresh read; if not, work the checklist yourself against the documents rather than from memory.

1. **Is it actually absent, or is it somewhere else?** Search `protocol.v1.json`, `cohort.json`, and all of `docs/` for the term and its synonyms before concluding it is missing. Several past blockers were answered in a file that had not been re-read since it changed.
2. **Does §3 resolve it?** If one branch is conservative and the other is not, you are done.
3. **Can it be derived?** From a cited standard, from an existing pattern in the same file, from arithmetic on values that do exist.
4. **Does it block *this step*, or a later one?** If later, note it and carry on. A missing value at Step 24 is not a Step 18 blocker.
5. **Would a wrong choice be caught downstream?** If a test at Step 11 or the design audit at Step 35 would catch it, take the safe branch now and let the test do its job.
6. **Would a competent engineer with clinical supervision ship a decision here without asking?** If yes, so should you.

If all six clear and it is still Tier 3, stop — and stop well (§7).

6. Reviewer subagent prompt

Hand this to a fresh subagent, along with the specific question and the relevant file excerpts. It has no stake in the answer and no memory of how long you have been staring at it.

You are reviewing whether a build agent's blocker is real or whether it is overthinking. You are not solving the problem.

Context: PatientTriage.ai, an emergency-department triage decision-support prototype. The build agent works from a complete specification set in docs/ plus two clinical data files, protocol.v1.json and cohort.json. Its standing instruction is: decide and proceed unless a wrong choice could change which triage band a patient is assigned AND there is no safe default. The safe default, always, is the more conservative option for the patient – higher acuity, wider uncertainty interval, more visible, more logged, abstain over assert.

The agent proposes to stop and escalate. Assess:

1. Is the information genuinely absent from the provided files, or did the agent miss it? Search the excerpts before answering.
2. Does the conservative default resolve it? If one branch is safer for the patient, say so and name it.
3. Can the value be derived from a cited standard, an existing pattern in the same file, or arithmetic on values that do exist? If so, show the derivation.
4. Does it block the CURRENT step, or a later one?
5. Would a competent engineer under clinical supervision ship a decision here without asking?

Return exactly one verdict:

PROCEED – with the specific decision to take and its one-line justification
 PROVISIONAL – with the safe branch to take and what to flag for later review
 ESCALATE – with the single sharpest question to ask, in one sentence

Bias toward PROCEED. The cost of a logged, reversible, conservative decision is much lower than the cost of a stalled build. Reserve ESCALATE for a clinical constant with no derivable basis, a missing safety gate, an absent content artefact, or an architectural property that makes a stated product claim unreachable.

7. How to stop, when you do stop

- **Batch.** Do not stop at the first Tier 3 in a phase. Continue on everything not downstream of it, collect the rest, and raise them together at the phase boundary. One message with four questions beats four messages.
- **Lead with what you already did.** State the steps completed and tests passing before the question.
- **Ask the sharpest version.** Not "how should paediatric SpO₂ be handled" but "paediatric SpO₂, supplemental oxygen and ACVPU have no defined Layer-1 contribution, and RULE-RESP-01 is adult-only, so a hypoxic child has no hard-rule gate."
- **Propose an answer.** Give the branch you would take under §3 and say you will proceed on it if there is no reply. Being wrong in a named direction is far more useful than being blocked in no direction.
- **Never idle.** Say what you are continuing with in the meantime.

8. What has not changed

The safe-default rule licenses decisions about the *specification*. It licenses nothing about the code.

- The eight clinical invariants in `AGENTS.md` are absolute. Never relax one to unblock yourself — model the case honestly instead, as `UNRESOLVABLE` did.
- The design contract in `04-UIUX-BRIEF.md` §2 is absolute.
- No build step, no framework, no runtime dependency.
- Clinical thresholds live in `protocol.v1.json`. A Tier 2 decision that needs a new constant **adds it to the protocol file**; it never hard-codes it.
- **Never edit a fixture, an `expect` block, or a golden file to make a test pass.** If one fails, the fixture and the code are competing claims about the same thing, and which is wrong is a decision — report it.
- Every decision is logged, whatever the tier. The log is the audit trail for the build, and it is the same argument the product makes about itself.